Final Assignment

ECON 452

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In [1]:
```
%pip install pyblp
```

```
Requirement already satisfied: pyblp in /opt/anaconda3/envs/marginaleffects/
lib/python3.13/site-packages (1.1.2)
Requirement already satisfied: numpy>=1.17.0 in /opt/anaconda3/envs/marginal
effects/lib/python3.13/site-packages (from pyblp) (2.3.4)
Requirement already satisfied: patsy>=0.5.1 in /opt/anaconda3/envs/marginale
ffects/lib/python3.13/site-packages (from pyblp) (1.0.2)
Requirement already satisfied: pyhdfe>=0.1.0 in /opt/anaconda3/envs/marginal
effects/lib/python3.13/site-packages (from pyblp) (0.2.0)
Requirement already satisfied: scipy>=1.2.0 in /opt/anaconda3/envs/marginale
ffects/lib/python3.13/site-packages (from pyblp) (1.16.2)
Requirement already satisfied: sympy>=1.1.0 in /opt/anaconda3/envs/marginale
ffects/lib/python3.13/site-packages (from pyblp) (1.14.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /opt/anaconda3/envs/mar
ginaleffects/lib/python3.13/site-packages (from sympy>=1.1.0->pyblp) (1.3.0)
Note: you may need to restart the kernel to use updated packages.
```

Model setup

We study demand and supply for pizza products in Westwood.

Demand:

$$u_{ijt} = \beta_i^p \times p_{jt} + \beta^d \times deliverytime_{jt} + \beta^r \times rating_{jt} + \xi_{jt} + \epsilon_{ijt}$$

$$u_{i0t} = \alpha_i + \epsilon_{i0t}$$

$$\beta_i^p = \beta^p + \pi \times \frac{1}{Income_i}$$

$$\alpha_i \sim N(0, \sigma^2)$$

Supply:

$$mc_{jt} = \gamma_0 + \gamma_1 \times rating_{jt} + \gamma_3 \times hourlywages_{jt} + \gamma_4 \times deliverytime_{jt} + \omega_{jt}$$

Firms compete in a differentiated-products Bertrand pricing game.

Question 1 Instrumental Variables

Questions:

1. **Explain why BLP instruments can be used to estimate demand parameters taking price endogeneity into account.**
2. **Explain verbally how `demand_instruments0` to `demand_instruments3` and `supply_instruments0` to `supply_instruments5` in `products_pizza.csv` are constructed.**
3. **Explain what shocks can be used as demand and supply instruments and how they relate to `demand_instruments4` and `supplyinstruments6`.**

Answer 1:

In the BLP demand model, price is endogenous because firms observe product characteristics (quality components) that the econometrician does not observe, and these unobserved quality components are captured by ξ_{jt}. If a product has higher unobserved quality, consumers value it more and firms can charge a higher price for it. As a result, price p_{jt} is correlated with ξ_{jt}, so directly regressing demand on price would produce a biased estimate of the price effect.

BLP instruments address this problem by using instrumental variables; variables that change equilibrium prices but do not directly affect utility from product j. This is the exclusion restriction: the instrument can affect demand for product j only indirectly through price, not by entering the utility of product j itself.

In this setting, the characteristics of other products sold by the same firm and of rival firms' products can affect substitution patterns, competitive pressure, and markups, so they help explain the equilibrium price of product j. But these other-product characteristics are not characteristics of product j itself, so they are excluded from the mean utility of product j. This gives the exogeneity condition needed for identification. After recovering mean utility δ_{jt}, the demand parameters are then identified using GMM moment conditions of the form $E[\xi_{jt} Z_{jt}] = 0$.

Answer 2:

The BLP instruments are built by adding up characteristics of other products in the same market, always excluding product j itself. There are two groups:

- non-rival goods: other pizzas sold by the same firm in the same market
- rival goods: pizzas sold by competing firms in the same market

So, the columns in the dataset can be interpreted as follows:

- demand_instruments0: sum of rating for the other pizzas sold by the same firm in the same market
- demand_instruments1: sum of rating for pizzas sold by rival firms in the same market
- demand_instruments2: sum of deliverytime for the other pizzas sold by the same firm in the same market
- demand_instruments3: sum of deliverytime for pizzas sold by rival firms in the same market

For the supply side, the same idea is used, but now the instruments are also built from variables that matter for costs:

- supply_instruments0: sum of hourlywages for the other pizzas sold by the same firm in the same market
- supply_instruments1: sum of hourlywages for pizzas sold by rival firms in the same market
- supply_instruments2: sum of rating for the other pizzas sold by the same firm in the same market
- supply_instruments3: sum of rating for pizzas sold by rival firms in the same market
- supply_instruments4: sum of deliverytime for the other pizzas sold by the same firm in the same market
- supply_instruments5: sum of deliverytime for pizzas sold by rival firms in the same market

The main idea is that characteristics of other pizzas affect the competitive environment around product j, so they can affect its equilibrium price. But because these sums are constructed from other products in the market rather than from product j itself, they do not directly enter the utility or marginal cost of product j. That is why they can be used as excluded instruments.

Answer 3:

Demand instruments can come from cost shocks or cost shifters, such as wages or other input costs. These variables affect equilibrium prices through marginal cost, but they do not directly enter consumer utility. In this dataset, demand_instruments4 corresponds to hourlywages, so it is a demand instrument because it shifts prices through firms' costs while being excluded from utility.

Supply instruments can come from demand shocks or demand shifters that affect demand but do not directly affect firms' marginal costs. Here, income fits that idea better than the product characteristics. Inverse income changes consumer price sensitivity, so it shifts demand, but it does not appear in the marginal cost equation.

The assignment appears to contain a typo when it refers to supplyinstruments6. In products_pizza.csv, the supply-instrument columns run only from supply_instruments0

to supply_instruments5, so there is no separate supply_instruments6 variable to interpret. The natural reading is that the question is asking for the supply-side analogue of demand_instruments4: a demand shifter that affects firms' pricing incentives but does not enter marginal cost directly. In this model, inverse income is the clearest example of that kind of supply instrument.

Question 2- model

Load Packages and Data

In [2]:
```
import pyblp
import numpy as np
import pandas as pd

# import os module
import os
import matplotlib.pyplot as plt

pyblp.options.digits = 2
pyblp.options.verbose = False
pyblp.__version__
```

Out[2]: '1.1.2'

In [3]:
```
product_data = pd.read_csv("products_pizza.csv")
product_data.head()
```

Out[3]:

	v1	market_ids	firm_ids	shares	demand_instruments0	demand_instruments1
0	0	0	0	0.006038	1.873110	9.154572
1	1	0	0	0.008047	2.242486	9.154572
2	2	0	0	0.004980	2.174373	9.154572
3	3	0	0	0.006846	2.471596	9.154572
4	4	0	0	0.011540	2.334877	9.154572

In [4]:
```
agent_data = pd.read_csv("agents_pizza.csv")
agent_data.head()
```

Out[4]:

	v1	market_ids	nodes0	nodes1	weights	invincome
0	0	0	0.052184	-0.127630	0.001	0.693219
1	1	0	-2.037323	-1.195798	0.001	0.177587
2	2	0	0.741513	0.443902	0.001	0.200069
3	3	0	-0.610369	-0.417627	0.001	0.664243
4	4	0	0.374030	-2.591600	0.001	0.307349

Clean data

In [5]:
```
product_data.isna().sum()
```

Out[5]:
```
v1                      0
market_ids              0
firm_ids                0
shares                  0
demand_instruments0     0
demand_instruments1     0
demand_instruments2     0
demand_instruments3     0
demand_instruments4     0
supply_instruments0     0
supply_instruments1     0
supply_instruments2     0
supply_instruments3     0
supply_instruments4     0
supply_instruments5     0
prices                  0
deliverytime          100
rating                  0
hourlywages             0
dtype: int64
```

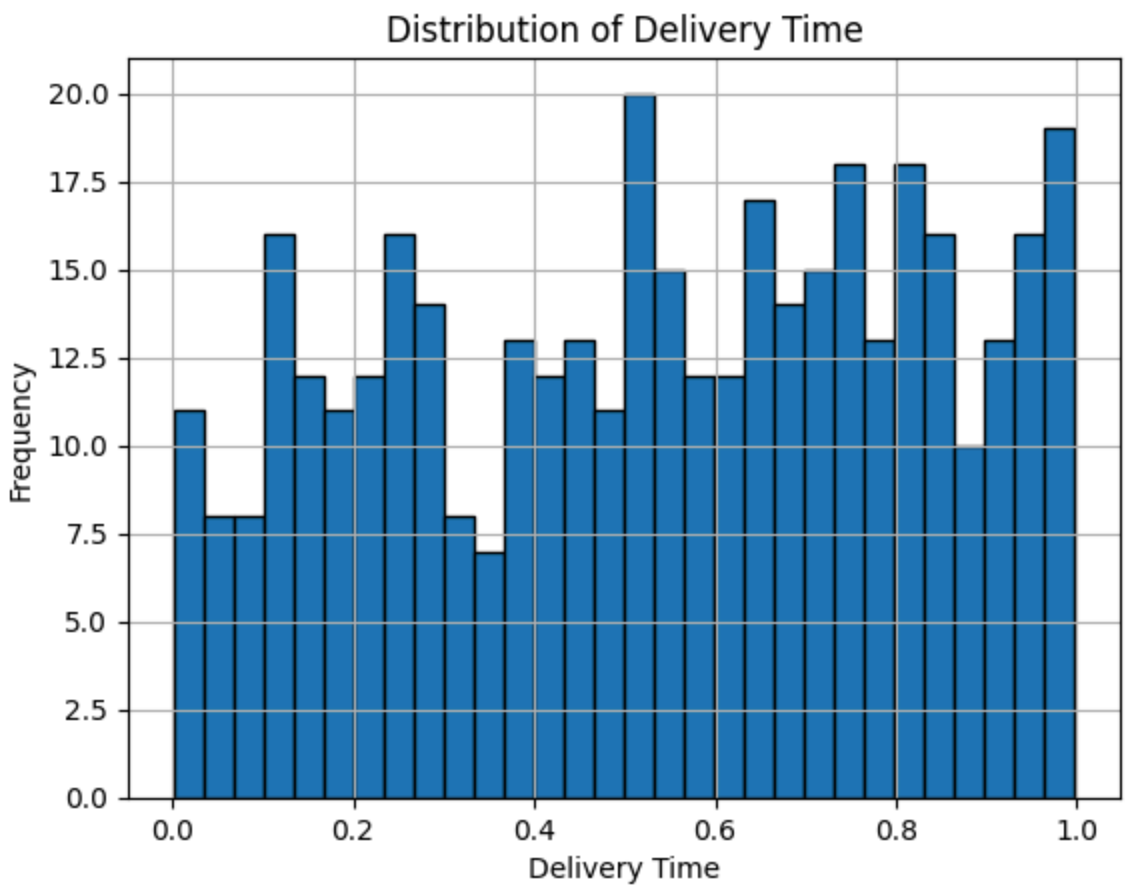
In [6]:
```
product_data.describe()
```

Out[6]:

	v1	market_ids	firm_ids	shares	demand_instruments0	dem
count	500.000000	500.000000	500.000000	500.000000	500.000000	
mean	249.500000	9.500000	2.400000	0.015191	1.624923	
std	144.481833	5.772056	1.745306	0.011910	0.525439	
min	0.000000	0.000000	0.000000	0.001773	0.261239	
25%	124.750000	4.750000	1.000000	0.006824	1.266643	
50%	249.500000	9.500000	2.000000	0.011253	1.640795	
75%	374.250000	14.250000	4.000000	0.020239	2.002215	
max	499.000000	19.000000	5.000000	0.068168	2.807821	

In [7]:

```
# check distribution of delivery time to decide on imputation
product_data['deliverytime'].dropna().hist(bins=30, edgecolor='black')
plt.xlabel('Delivery Time')
plt.ylabel('Frequency')
plt.title('Distribution of Delivery Time')
plt.show()
```



In [8]:

```
# # mean impute null values in delivery time
# product_data['deliverytime'].fillna(product_data['deliverytime'].mean(), i
```

In [9]:
```
# Mean imputation by product / firm pair
# Step 2: Implement group-based mean imputation
# Calculate the mean deliverytime for each (market_ids, firm_ids) group
group_means = product_data.groupby(['market_ids', 'firm_ids'])['deliverytime

# Create a copy to impute missing values
product_data_imputed = product_data.copy()

# Impute missing values with group means
missing_mask = product_data_imputed['deliverytime'].isna()
product_data_imputed.loc[missing_mask, 'deliverytime'] = group_means[missing

# override original
product_data = product_data_imputed
```

In [10]:
```
agent_data.isna().sum()
```

Out[10]:
```
v1            0
market_ids    0
nodes0        0
nodes1        0
weights       0
invincome     0
dtype: int64
```

Set up the problem

In [11]:
```
# set up the problem
product_formulations = (
   pyblp.Formulation('0 + prices + deliverytime + rating'),
   pyblp.Formulation('1 + prices'),
   pyblp.Formulation('1 + rating + hourlywages + deliverytime')
)
product_formulations
```

Out[11]:
```
(prices + deliverytime + rating,
 1 + prices,
 1 + rating + hourlywages + deliverytime)
```

In [12]:
```
# set up the problem
agent_formulation = pyblp.Formulation('0 + invincome')
agent_formulation
```

Out[12]:
```
invincome
```

Initialize the problem

In [13]:
```
# initialize the problem
problem = pyblp.Problem(product_formulations, product_data, agent_formulatic
problem
```

Out[13]:
```
Dimensions:
=========================================================
 T    N    F     I     K1    K2    K3    D    MD    MS
---  ---  ---  -----  ----  ----  ----  ---  ----  ----
20   500   6   20000   3     2     4     1    7     10
=========================================================

Formulations:
==============================================================================
========
        Column Indices:              0           1             2
3
-------------------------------  ---------  ------------  -----------  ----
--------
  X1: Linear Characteristics      prices    deliverytime    rating
 X2: Nonlinear Characteristics       1         prices
X3: Linear Cost Characteristics      1         rating     hourlywages  deli
verytime
        d: Demographics          invincome
==============================================================================
========
```

Solving the problem

In [14]:
```
# set starting values
initial_sigma = np.diag([1, 0])
initial_pi = np.c_[[0, -1]]
initial_beta=np.c_[[-1, -1, 1]]
initial_gamma=np.c_[[-1, 1, 1, -1]]
```

In [15]:
```
results = problem.solve(
    initial_sigma,
    initial_pi,
    beta=initial_beta,
    gamma=initial_gamma,
    costs_bounds=(0.001, None),
    #W_type='clustered',
    #se_type='clustered',
    initial_update=True,
    optimization=pyblp.Optimization('l-bfgs-b', {'gtol': 1e-5})
)
results
```

Out[15]:
```
Problem Results Summary:
==============================================================================
=============================================
GMM   Objective    Projected    Reduced Hessian  Reduced Hessian  Clipped
Clipped  Weighting Matrix  Covariance Matrix
Step    Value    Gradient Norm  Min Eigenvalue   Max Eigenvalue   Shares
Costs   Condition Number  Condition Number
----  ---------  -------------  ---------------  ---------------  -------
-------  ----------------  -----------------
 2    +1.8E+01     +7.0E-03        +5.5E+00         +2.7E+05         0
0         +1.2E+05          +5.8E+05
==============================================================================
=============================================

Cumulative Statistics:
==============================================================================
Computation  Optimizer  Optimization   Objective   Fixed Point  Contraction
   Time      Converged   Iterations   Evaluations  Iterations   Evaluations
-----------  ---------  ------------  -----------  -----------  -----------
 00:00:38       Yes         298           352         50650       157300
==============================================================================

Nonlinear Coefficient Estimates (Robust SEs in Parentheses):
=====================================================
Sigma:      1        prices   |   Pi:    invincome
------  ----------  --------  |  ------  ----------
  1      +1.3E-01             |    1      +0.0E+00
        (+2.1E+00)            |
                              |
prices   +0.0E+00   +0.0E+00  |  prices   -4.1E+00
                              |          (+8.2E-02)
=====================================================

Beta Estimates (Robust SEs in Parentheses):
=====================================
  prices    deliverytime    rating
----------  ------------  ----------
 -1.1E-02     -8.1E-01     +9.0E-01
(+1.5E-02)   (+6.3E-02)   (+3.8E-02)
=====================================

Gamma Estimates (Robust SEs in Parentheses):
====================================================
    1         rating    hourlywages  deliverytime
----------  ----------  -----------  ------------
 +6.9E-01    +4.7E-01    +1.4E+00      -6.7E-01
(+6.0E-02)  (+3.6E-02)  (+8.1E-02)    (+6.0E-02)
====================================================
```

Interpretation

- linear demand (β): The linear demand estimates imply that delivery time has a coefficient of about -0.85 and rating has a coefficient of about 0.92. So a one-unit increase in delivery time lowers mean utility by about 0.85, while a one-unit increase

in rating raises mean utility by about 0.92. This is consistent with the idea that consumers prefer faster delivery and better quality.

- marginal cost (γ) The marginal cost estimates imply that the constant is about 0.69, the coefficient on rating is about 0.47, the coefficient on hourly wages is about 1.3, and the coefficient on delivery time is about -0.69. So higher ratings and higher wages are associated with higher marginal costs, while longer delivery times are associated with lower marginal cost. The wage effect is intuitive because labor is a direct input into production and delivery. The negative coefficient on delivery time suggests that slower delivery may reflect lower-cost production or staffing choices.

- nonlinear estimates (σ) and (π) The nonlinear part allows heterogeneity in the utility of the outside good and in price sensitivity through inverse income. The estimated demographic interaction π is about -4.2, which means inverse income has a strong effect on price sensitivity: consumers with lower inverse income are less price sensitive, while consumers with higher inverse income are more price sensitive.

- starting values We chose starting values to match the signs we expected economically and to keep the optimization numerically stable. We started σ at `diag([1, 0])` because the outside-good random coefficient should have some dispersion, while the price coefficient does not have its own random term in our specification. We set $\pi = [0, -1]$ so that the demographic interaction starts from the idea that higher inverse income makes consumers more price sensitive. We initialized $\beta = [-1, -1, 1]$ so price and delivery time begin with negative effects on utility and rating begins with a positive effect. We initialized $\gamma = [-1, 1, 1, -1]$ so the cost function starts from the idea that ratings and wages raise cost, while longer delivery times are associated with lower marginal cost. These are not imposed restrictions on the final estimates; they are just reasonable starting points that help the solver converge to an economically sensible solution.

Question 3 - Elasticities and Diversion ratios

In [16]:
```
elasticities = results.compute_elasticities()
diversions = results.compute_diversion_ratios()
```

Elasticities

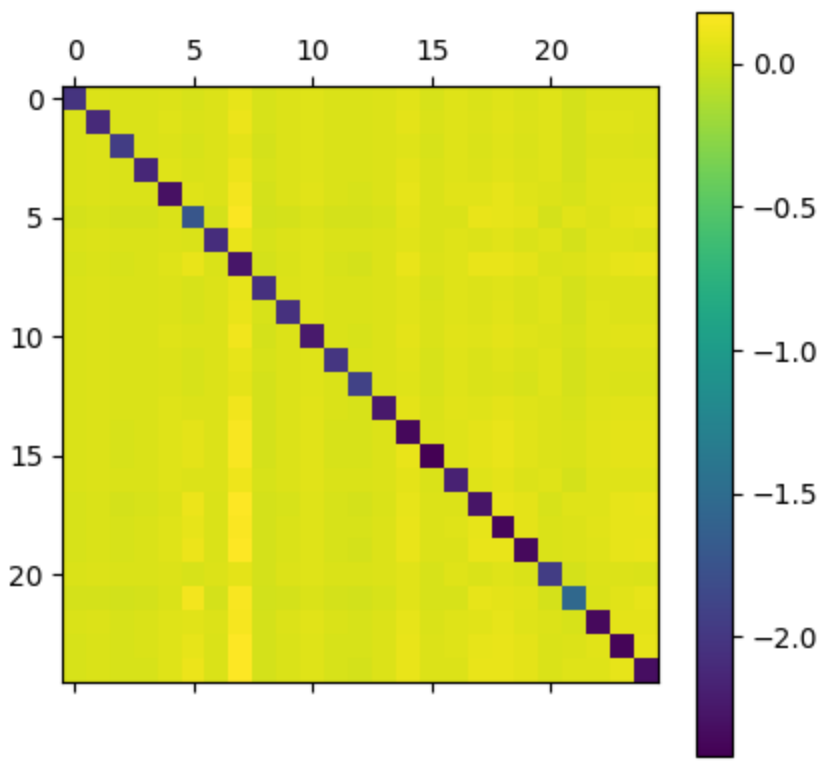
In [17]:
```
# look at market ids
product_data['market_ids'].unique()
```

Out[17]:
```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
       17, 18, 19])
```

In [18]:
```
product_data.groupby('market_ids')['shares'].sum()
```

Out[18]:
```
market_ids
0     0.424036
1     0.349357
2     0.387672
3     0.388787
4     0.389075
5     0.390935
6     0.363258
7     0.409922
8     0.364449
9     0.388503
10    0.366352
11    0.376966
12    0.329694
13    0.413926
14    0.327866
15    0.394372
16    0.363029
17    0.415907
18    0.388556
19    0.362657
Name: shares, dtype: float64
```

In [19]:
```
# Plot elasticities for market 0
single_market = product_data['market_ids'] == 0
plt.colorbar(plt.matshow(elasticities[single_market]));
```



Interpretation

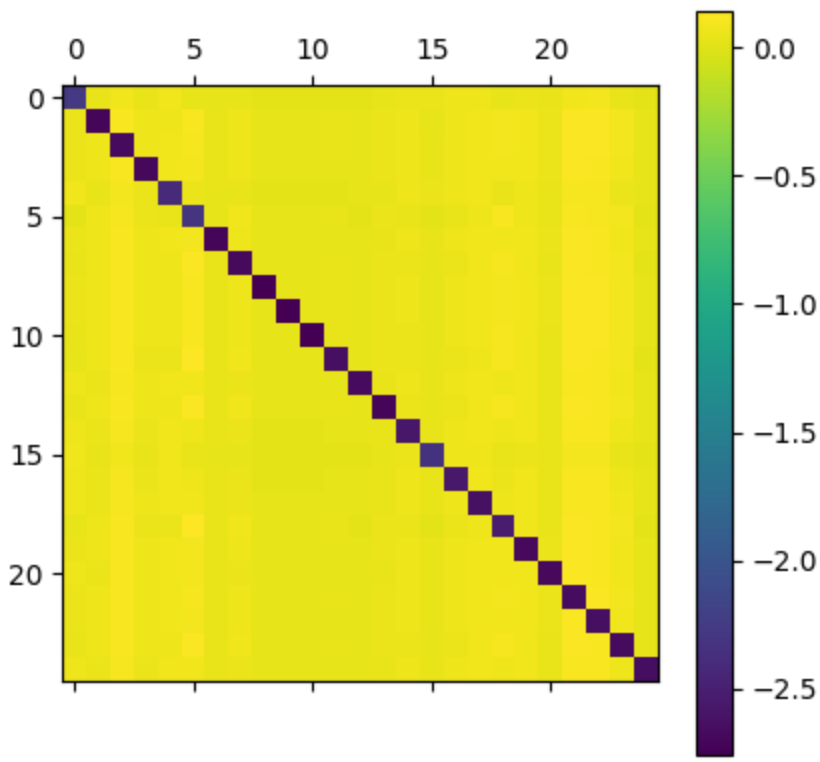
The own-price elasticities are on the diagonal, and they look to be around -1.5 to -2.0 in this market. That means demand is elastic: a 1 percent increase in price leads to a more

than 1 percent drop in quantity demanded.

The cross-price elasticities are the off-diagonal entries. They are small and close to 0, which suggests that when one pizza place raises its price, demand for the other pizzas changes only a little.

So overall, consumers seem quite responsive to a product's own price, but substitution across specific pizza options is fairly limited in this market.

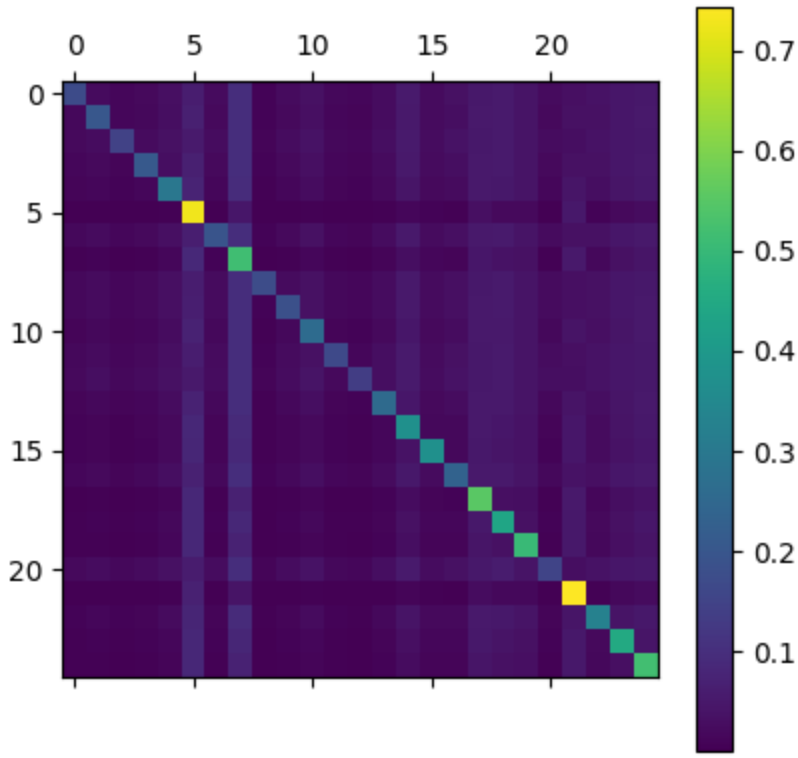
In [20]:
```
# Plot elasticities for market 14
single_market2 = product_data['market_ids'] == 14
plt.colorbar(plt.matshow(elasticities[single_market2]));
```



Interpretation

The first market I plotted was market 0. I also plotted market 14, which has a relatively low inside share. In market 14 the own-price elasticities are larger in absolute value, while the cross-price elasticities remain close to 0. This is consistent with the logit demand structure, where products with smaller shares tend to have more elastic demand because consumers can more easily substitute away from them. The cross-price elasticities being near 0 suggests that when a product raises its price, much of the substitution goes to the outside option rather than to a specific competing pizza.

In [21]:
```
# Plotting diversion ratios
plt.colorbar(plt.matshow(diversions[single_market]));
```



Interpretation

The off-diagonal entries are diversion to other pizza places, and most of them look to be around 10 to 20 percent. So if one pizza place raises its price, some consumers switch to competing pizza options, but not a huge share.

The diagonal entries capture diversion to the outside good, and they are much larger, around 30 to 70 percent. That means a lot of consumers respond to a price increase by leaving the pizza market rather than switching to a different pizza place.

This suggests that pizza places do compete with one another, but a large part of substitution is still between buying pizza and choosing the outside option.

Question 4 - Marginal Costs, Markups, HHI, Consumer Surplus, and Profits

In [22]:
```
# Question 4: model-based objects from the estimated equilibrium.

# Compute marginal costs, markups, and HHI from the estimated pyblp results.
q4_costs = results.compute_costs().flatten()
q4_markups = results.compute_markups(costs=q4_costs).flatten()
q4_hhi = results.compute_hhi().flatten()

# Scale profits and consumer surplus using the assignment's market size and
q4_market_population = 100_000
q4_slices_per_pizza = 6
```

```
q4_profits = results.compute_profits(costs=q4_costs).flatten() * q4_market_p
q4_cs = results.compute_consumer_surpluses().flatten() * q4_market_populatic

# Build a product-level table with baseline outcomes and the model-implied c
q4_product_table = product_data[['v1', 'market_ids', 'firm_ids', 'prices', '
q4_product_table['marginal_cost'] = q4_costs
q4_product_table['markup'] = q4_markups
q4_product_table['profit'] = q4_profits

# Build market-level summaries for concentration and consumer surplus, then
q4_market_table = pd.DataFrame({
    'market_ids': pd.Index(sorted(product_data['market_ids'].unique())),
    'hhi': q4_hhi,
    'consumer_surplus': q4_cs,
})

q4_mc_summary = pd.Series(q4_costs).describe().to_frame('marginal_cost')
q4_markup_summary = pd.Series(q4_markups).describe().to_frame('markup')
q4_hhi_summary = pd.Series(q4_hhi).describe().to_frame('hhi')
q4_profit_summary = pd.Series(q4_profits).describe().to_frame('profit')
q4_cs_summary = pd.Series(q4_cs).describe().to_frame('consumer_surplus')
```

In [23]:

```
display(q4_product_table.head(10))
display(q4_market_table.head(10))
display(q4_mc_summary)
display(q4_markup_summary)
display(q4_hhi_summary)
display(q4_profit_summary)
display(q4_cs_summary)
print(f"Markets with HHI >= 2500: {(q4_market_table['hhi'] >= 2500).sum()} c

fig, axes = plt.subplots(2, 3, figsize=(15, 8))
axes = axes.flatten()
plot_data = [
    (q4_costs, 'Marginal Costs', 'Marginal Cost'),
    (q4_markups, 'Markups', 'Markup'),
    (q4_hhi, 'HHI Across Markets', 'HHI'),
    (q4_profits, 'Profits Across Products', 'Profit'),
    (q4_cs, 'Consumer Surplus Across Markets', 'Consumer Surplus'),
]
for ax, (data, title, xlabel) in zip(axes, plot_data):
    ax.hist(data, bins=30, edgecolor='black')
    ax.set_title(title)
    ax.set_xlabel(xlabel)
    ax.set_ylabel('Count')
axes[-1].axis('off')
plt.tight_layout()
plt.show()
```

	v1	market_ids	firm_ids	prices	shares	marginal_cost	markup	prof
0	0	0	0	4.586001	0.006038	2.138679	0.533651	8865.58036
1	1	0	0	4.171702	0.008047	2.037529	0.511583	10304.27762
2	2	0	0	4.966030	0.004980	2.219800	0.553003	8206.07829
3	3	0	0	4.120471	0.006846	2.023821	0.508837	8611.70513
4	4	0	0	3.267534	0.011540	1.744537	0.466100	10544.77864
5	5	0	1	0.968238	0.060462	0.353261	0.635151	22309.65107
6	6	0	1	4.306469	0.007761	2.078307	0.517399	10376.18374
7	7	0	1	1.987181	0.050265	1.034755	0.479285	28724.3610
8	8	0	1	4.546549	0.003045	2.146853	0.527806	4384.2016
9	9	0	2	4.438455	0.007630	2.137193	0.518483	10535.44062

	market_ids	hhi	consumer_surplus
0	0	2040.019633	321607.791000
1	1	2266.968603	440269.937928
2	2	1910.292655	326029.759978
3	3	1705.866064	396284.890999
4	4	1810.765776	433376.015985
5	5	1713.712189	501164.782410
6	6	1768.664444	346086.532626
7	7	1824.744946	422762.057109
8	8	1914.131328	363761.890344
9	9	1727.717476	338896.526215

	marginal_cost
count	500.000000
mean	1.823218
std	0.578773
min	0.230516
25%	1.429346
50%	1.845768
75%	2.246162
max	3.166607

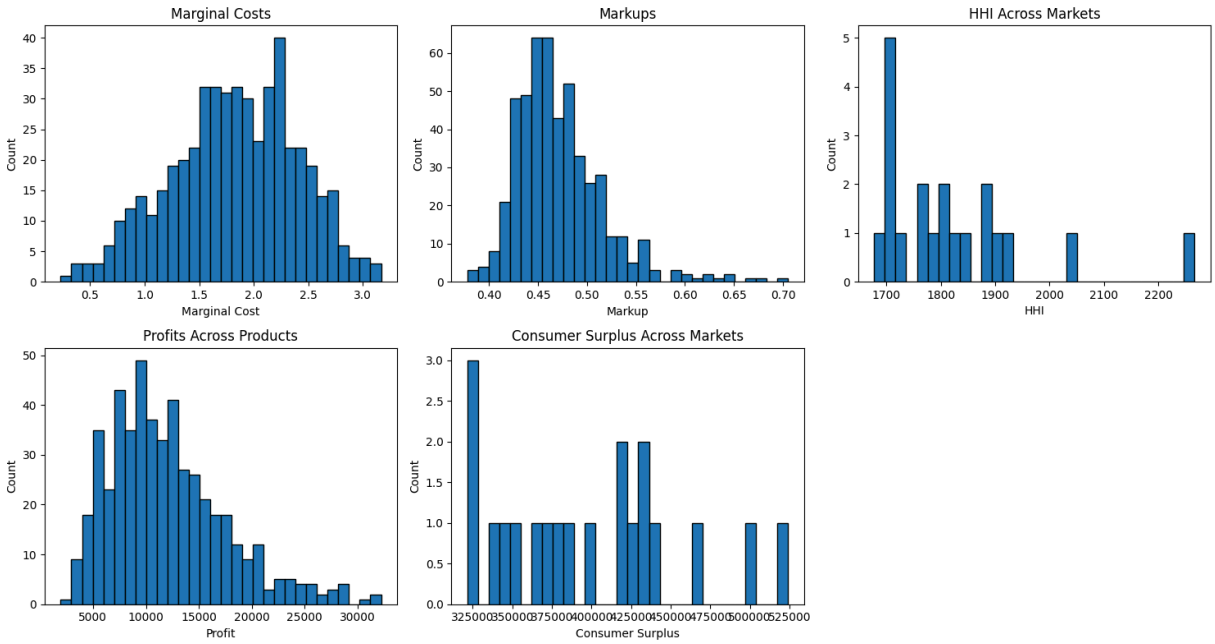
	markup
count	500.000000
mean	0.472042
std	0.045229
min	0.377983
25%	0.441525
50%	0.462936
75%	0.493525
max	0.705023

	hhi
count	20.000000
mean	1823.348699
std	140.186253
min	1677.246362
25%	1712.707151
50%	1795.216258
75%	1886.604176
max	2266.968603

	profit
count	500.000000
mean	12007.527552
std	5547.711984
min	1948.560693
25%	7923.397612
50%	11020.381518
75%	14907.541818
max	32211.083933

	consumer_surplus
count	20.000000
mean	399022.210219
std	58108.069800
min	321607.791000
25%	352899.279716
50%	392503.854908
75%	432507.001063
max	524058.019580

```
Markets with HHI >= 2500: 0 out of 20
```



Interpretation

The average marginal cost is about 1.82 dollars per slice. Since average prices in the sample are around 3 to 5 dollars per slice, this suggests that marginal cost is a substantial share of price, but still leaves room for positive markups.

The average markup is about 0.47, so on average firms keep roughly 47 percent of price above marginal cost. This is economically large but still plausible in a differentiated-products market. Using the Lerner-index logic, a markup of this size is broadly consistent with the elasticities we found above: with average own-price elasticities around 1.5 to 2 in absolute value, firms can sustain meaningful but not extreme price-cost margins.

The average HHI is about 1823, and no market has HHI above 2500. So none of the markets would be classified as highly concentrated under the usual threshold.

Assuming a full pizza has 6 slices and there are 100,000 consumers in each market, the average profit is about 12,008 dollars per product.

Under the same scaling, the average consumer surplus is about 399,022 dollars per market.

Question 5 - Merger Analysis

In [24]:
```
# Question 5: merger counterfactual using Week 10 methods.

# Reassign ownership so firms 0, 1, 2, and 3 are treated as one merged firm
q5_merger_ids = product_data['firm_ids'].replace({0: 1, 2: 1, 3: 1}).to_nump

# Use 100,000 consumers in total across markets for the merger consumer-surp
q5_market_population = 100_000 / product_data['market_ids'].nunique()

# Solve for the post-merger equilibrium and compute the new prices, shares,
q5_changed_prices = results.compute_prices(firm_ids=q5_merger_ids, costs=q4_
q5_changed_shares = results.compute_shares(q5_changed_prices).flatten()
q5_changed_hhi = results.compute_hhi(firm_ids=q5_merger_ids, shares=q5_chang
q5_changed_markups = results.compute_markups(q5_changed_prices, q4_costs).fl
q5_changed_profits = results.compute_profits(q5_changed_prices, q5_changed_s
q5_changed_cs = results.compute_consumer_surpluses(q5_changed_prices).flatte

# Build a product-level table that compares baseline and post-merger outcome
q5_product_table = product_data[['v1', 'market_ids', 'firm_ids', 'prices', '
q5_product_table['post_merger_firm_id'] = q5_merger_ids
q5_product_table['new_price'] = q5_changed_prices
q5_product_table['new_share'] = q5_changed_shares
q5_product_table['delta_price'] = q5_product_table['new_price'] - q5_product
q5_product_table['delta_share'] = q5_product_table['new_share'] - q5_product
q5_product_table['baseline_markup'] = q4_markups
q5_product_table['new_markup'] = q5_changed_markups
q5_product_table['delta_markup'] = q5_product_table['new_markup'] - q5_produ
q5_product_table['baseline_profit'] = q4_profits
q5_product_table['new_profit'] = q5_changed_profits
q5_product_table['delta_profit'] = q5_product_table['new_profit'] - q5_produ
```

In [25]:
```
# Build a market-level table that compares baseline and post-merger HHI and
q5_market_table = pd.DataFrame({
    'market_ids': pd.Index(sorted(product_data['market_ids'].unique())),
    'baseline_hhi': q4_hhi,
    'new_hhi': q5_changed_hhi,
    'delta_hhi': q5_changed_hhi - q4_hhi,
    'baseline_cs': q4_cs / product_data['market_ids'].nunique(),
    'new_cs': q5_changed_cs,
    'delta_cs': q5_changed_cs - (q4_cs / product_data['market_ids'].nunique(
})
```

In [26]:
```
# Display the main product-level and market-level outputs together with desc
display(q5_product_table.head(10))
display(q5_market_table.head(10))
```

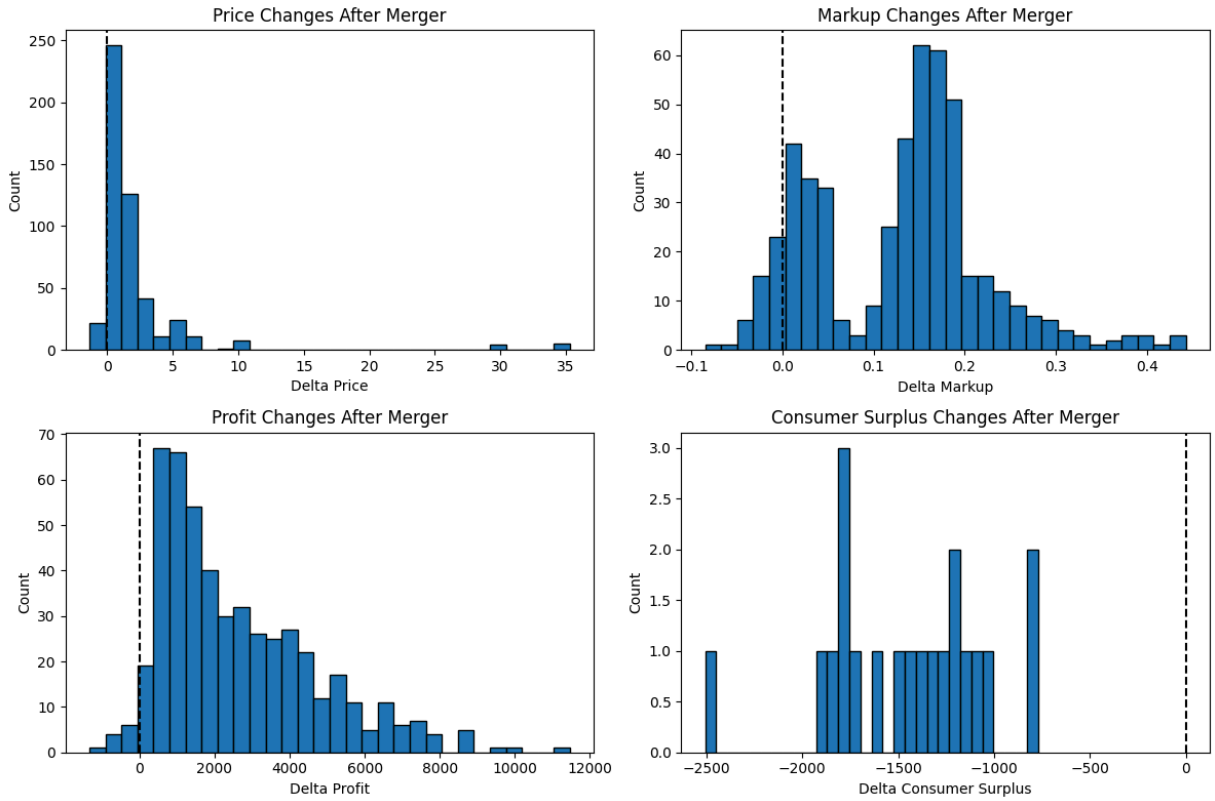
```
display(q5_product_table[['delta_price', 'delta_share', 'delta_markup', 'del
display(q5_market_table[['delta_hhi', 'delta_cs']].describe())

# Plot the distributions of merger-induced changes in prices, markups, profi
fig, axes = plt.subplots(2, 2, figsize=(12, 8))
plots = [
    (q5_product_table['delta_price'], 'Price Changes After Merger', 'Delta F
    (q5_product_table['delta_markup'], 'Markup Changes After Merger', 'Delta
    (q5_product_table['delta_profit'], 'Profit Changes After Merger', 'Delta
    (q5_market_table['delta_cs'], 'Consumer Surplus Changes After Merger', '
]
for ax, (series, title, xlabel) in zip(axes.flatten(), plots):
    ax.hist(series, bins=30, edgecolor='black')
    ax.axvline(0, linestyle='--', color='black')
    ax.set_title(title)
    ax.set_xlabel(xlabel)
    ax.set_ylabel('Count')
plt.tight_layout()
plt.show()
```

	v1	market_ids	firm_ids	prices	shares	post_merger_firm_id	new_price	new
0	0	0	0	4.586001	0.006038	1	9.568518	0.
1	1	0	0	4.171702	0.008047	1	9.170513	0.
2	2	0	0	4.966030	0.004980	1	9.862617	0.
3	3	0	0	4.120471	0.006846	1	9.113322	0.
4	4	0	0	3.267534	0.011540	1	7.651632	0.
5	5	0	1	0.968238	0.060462	1	1.094962	0.
6	6	0	1	4.306469	0.007761	1	9.335763	0.
7	7	0	1	1.987181	0.050265	1	2.502802	0.
8	8	0	1	4.546549	0.003045	1	9.599060	0
9	9	0	2	4.438455	0.007630	1	9.562944	0.

	market_ids	baseline_hhi	new_hhi	delta_hhi	baseline_cs	new_cs
0	0	2040.019633	3467.685339	1427.665706	16080.389550	14155.314159
1	1	2266.968603	4612.916505	2345.947902	22013.496896	20759.132895
2	2	1910.292655	4552.501071	2642.208416	16301.487999	15208.238622
3	3	1705.866064	3908.481412	2202.615349	19814.244550	18660.458011
4	4	1810.765776	4346.383400	2535.617625	21668.800799	20241.156230
5	5	1713.712189	3665.181943	1951.469753	25058.239120	23677.373825
6	6	1768.664444	3930.406485	2161.742041	17304.326631	16087.985566
7	7	1824.744946	3690.212648	1865.467702	21138.102855	19420.544839
8	8	1914.131328	3713.913731	1799.782403	18188.094517	16848.071795
9	9	1727.717476	3857.315845	2129.598369	16944.826311	16135.167783

	delta_price	delta_share	delta_markup	delta_profit
count	500.000000	500.000000	500.000000	500.000000
mean	2.060365	-0.002800	0.129267	2607.945951
std	4.577638	0.005255	0.095302	2095.027451
min	-1.362391	-0.017323	-0.084898	-1334.502608
25%	0.181829	-0.006308	0.038875	987.296933
50%	0.996717	-0.003464	0.145669	1998.098353
75%	2.059851	0.002449	0.181594	3867.940827
max	35.322241	0.009157	0.442877	11469.714184

	delta_hhi	delta_cs
count	20.000000	20.000000
mean	2037.931751	-1457.333470
std	296.990657	420.670346
min	1427.665706	-2506.215907
25%	1869.847462	-1756.347238
50%	1982.154641	-1404.254932
75%	2215.045235	-1200.702434
max	2642.208416	-771.756421



Interpretation

After the merger, prices increase on average and market shares fall on average for individual products. The average change in HHI is about 2,038 points across markets, which is a very large increase in concentration.

Markups increase on average by about 0.129 across products. This makes sense because once firms 0, 1, 2, and 3 act as one owner, they internalize competition across their products and have a stronger incentive to raise prices.

Profits increase on average by about 2,608 dollars per product. The increase in profits is driven by the higher post-merger prices and markups, even though some products lose market share.

Consumer surplus decreases in every market. The average change is about -1,457 dollars per market when total population is fixed at 100,000 consumers across all markets. This happens because the merger raises concentration and prices, which makes consumers worse off on average.

Question 6 Delivery Ceiling

- we return to the original 6-firm market structure (question 2)
- cap delivery time at 33 minutes (33/60 hours),

- hold estimated unobserved demand shocks ξ and cost shocks ω fixed,
- initialize a pyblp.Simulation, and
- solve for the new endogenous equilibrium with replace_endogenous().

In [27]:
```
# Start from the same cleaned data and formulations used in Question 2.
q6_products = product_data.copy()
q6_agents = agent_data.copy()
q6_products['deliverytime_binding'] = (q6_products['deliverytime'] > 33 / 60
q6_products['deliverytime'] = q6_products['deliverytime'].clip(upper=33 / 60

# Hold estimated unobservables fixed, following Week 10.
xi = results.xi
omega = results.omega

q6_simulation = pyblp.Simulation(
    product_formulations=product_formulations,
    beta=results.beta,
    sigma=results.sigma,
    gamma=results.gamma,
    product_data=q6_products.drop(columns=["deliverytime_binding"]),
    agent_data=q6_agents,
    agent_formulation=agent_formulation,
    pi=results.pi,
    xi=xi,
    omega=omega,
    seed=1,
)

# Replace simulated prices and market shares with equilibrium values consist
q6_simulation_results = q6_simulation.replace_endogenous()
q6_simulation_results
```

Out[27]:
```
Simulation Results Summary:
==============================================================================
===========================
Computation  Fixed Point  Fixed Point  Contraction  Profit Gradients  Profi
t Hessians  Profit Hessians
   Time       Failures    Iterations   Evaluations      Max Norm      Min E
igenvalue   Max Eigenvalue
-----------  -----------  -----------  -----------  ----------------  -----
----------  ---------------
 00:00:00         0          1519         1519          +8.4E-13         -
9.1E-02         -1.3E-05
==============================================================================
===========================
```

6.1. New equilibrium prices and market shares

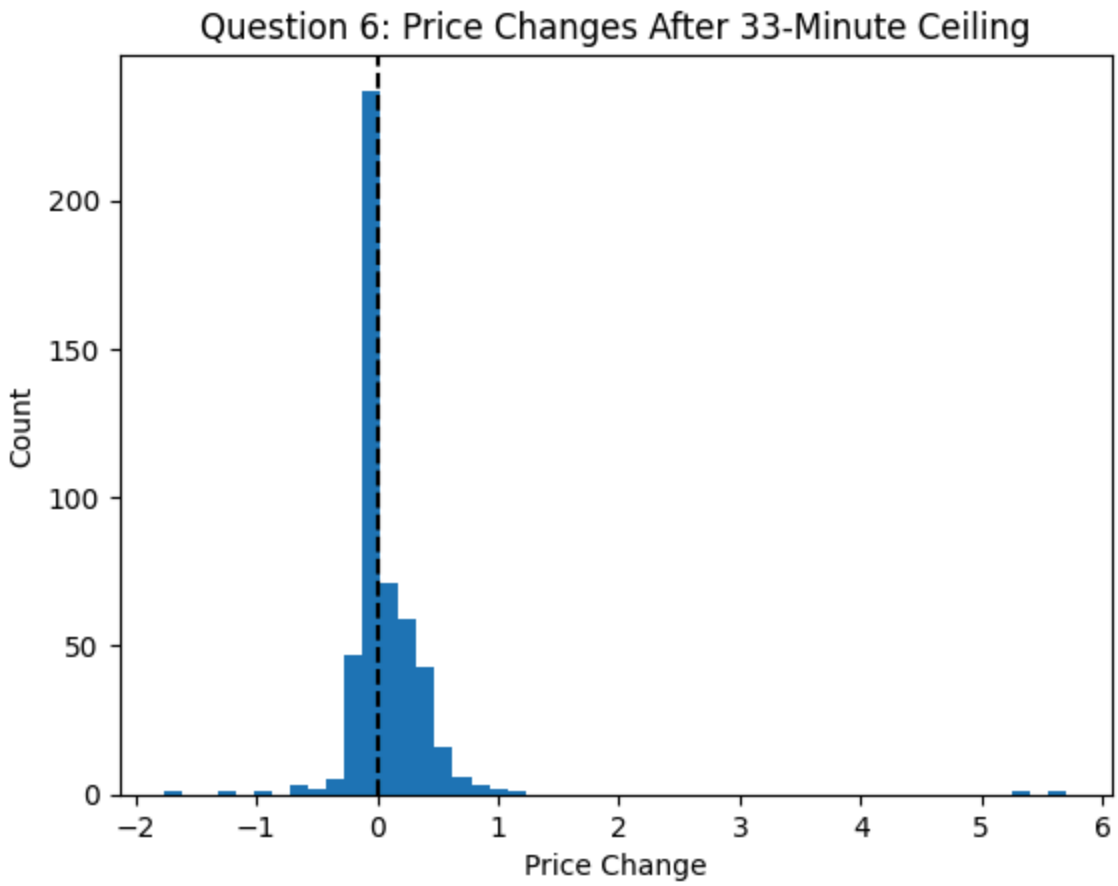
In [28]:
```
# Convert solved simulation into product data, following Week 10.
q6_problem = q6_simulation_results.to_problem()
q6_product_data_new = q6_problem.products
q6_df = pd.DataFrame(pyblp.data_to_dict(q6_product_data_new))

q6_equilibrium = product_data[['v1', 'market_ids', 'firm_ids', 'prices', 'sh
```

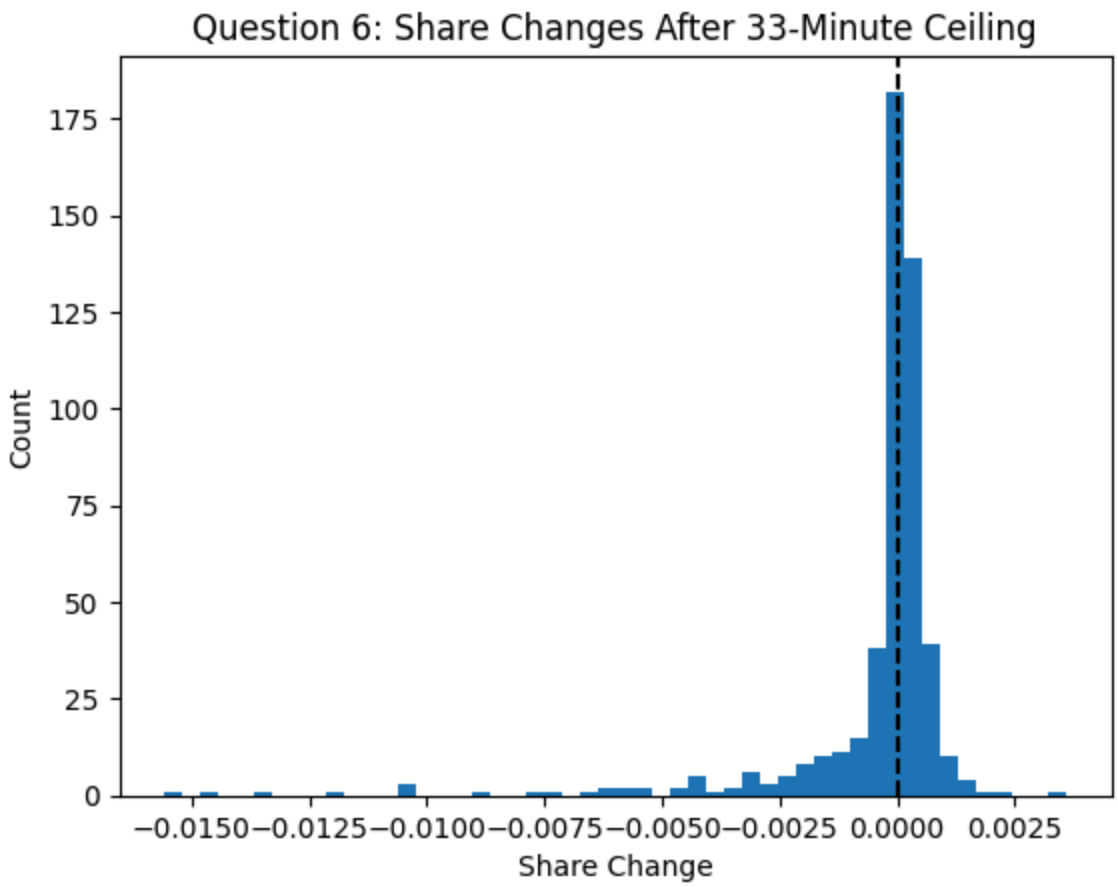
```
q6_equilibrium = q6_equilibrium.rename(columns={'prices': 'baseline_price',
q6_equilibrium['new_price'] = q6_df['prices'].to_numpy()
q6_equilibrium['new_share'] = q6_df['shares'].to_numpy()
q6_equilibrium['delta_price'] = q6_equilibrium['new_price'] - q6_equilibrium
q6_equilibrium['delta_share'] = q6_equilibrium['new_share'] - q6_equilibrium

price_change = q6_df["prices"] - product_data["prices"]
share_change = q6_df["shares"] - product_data["shares"]

display(q6_equilibrium.head(10))
display(pd.Series(price_change).describe().to_frame("delta_price"))
display(pd.Series(share_change).describe().to_frame("delta_share"))

plt.hist(price_change, bins=50)
plt.axvline(0, linestyle="--", color="black")
plt.title("Question 6: Price Changes After 33-Minute Ceiling")
plt.xlabel("Price Change")
plt.ylabel("Count")
plt.show()

plt.hist(share_change, bins=50)
plt.axvline(0, linestyle="--", color="black")
plt.title("Question 6: Share Changes After 33-Minute Ceiling")
plt.xlabel("Share Change")
plt.ylabel("Count")
plt.show()
```

	v1	market_ids	firm_ids	baseline_price	baseline_share	new_price	new_share	del
0	0	0	0	4.586001	0.006038	4.345851	0.006464	-(
1	1	0	0	4.171702	0.008047	4.203285	0.008189	(
2	2	0	0	4.966030	0.004980	4.674701	0.005363	-(
3	3	0	0	4.120471	0.006846	3.942988	0.007251	-(
4	4	0	0	3.267534	0.011540	3.179773	0.011991	-(
5	5	0	1	0.968238	0.060462	0.956670	0.062860	-
6	6	0	1	4.306469	0.007761	4.177930	0.007941	-(
7	7	0	1	1.987181	0.050265	2.434576	0.042405	(
8	8	0	1	4.546549	0.003045	4.402953	0.003114	-(
9	9	0	2	4.438455	0.007630	4.832485	0.007242	(

	delta_price
count	500.000000
mean	0.080929
std	0.428287
min	-1.762756
25%	-0.074958
50%	-0.014041
75%	0.193194
max	5.702439

	delta_share
count	500.000000
mean	-0.000419
std	0.001951
min	-0.015580
25%	-0.000229
50%	0.000067
75%	0.000289
max	0.003595

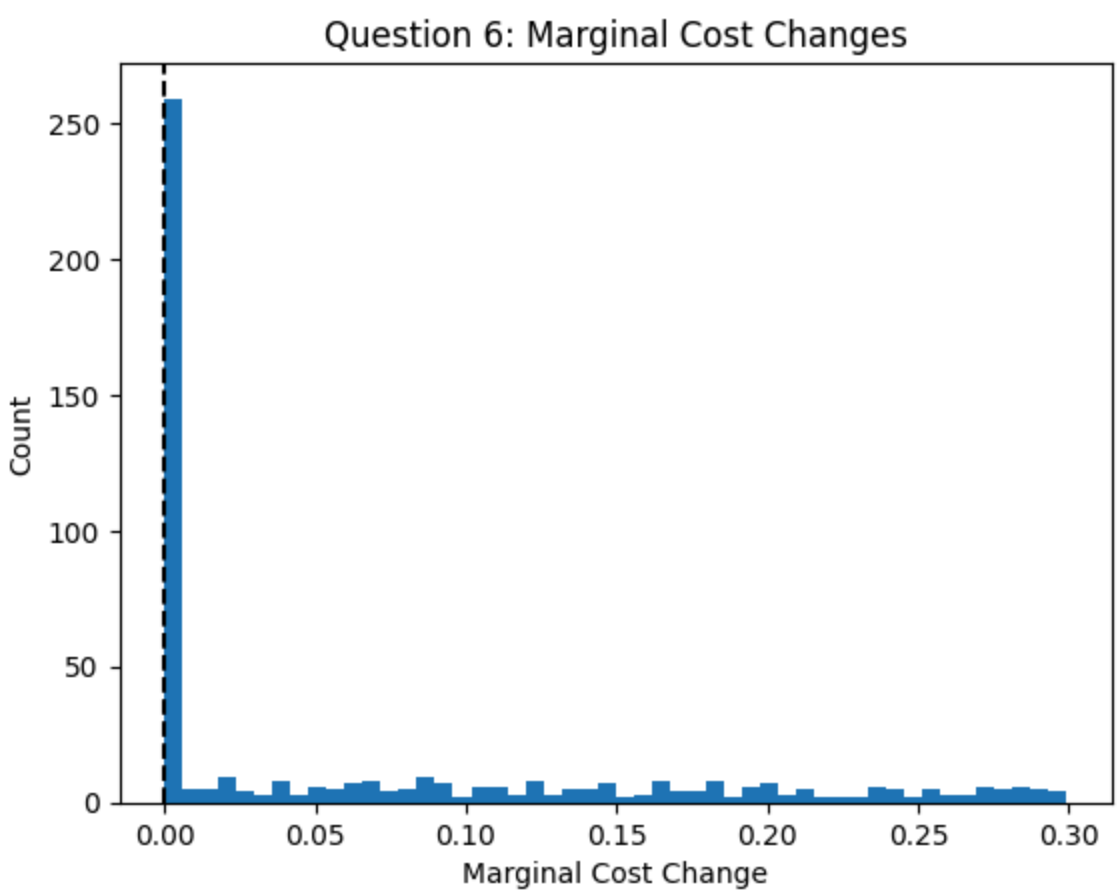




6.2. New marginal costs

In [29]:
```
mc_q6 = q6_simulation_results.compute_costs().flatten()
mc_orig = results.compute_costs().flatten()
mc_change = mc_q6 - mc_orig

display(pd.Series(mc_change).describe().to_frame("delta_mc"))
plt.hist(mc_change, bins=50)
plt.axvline(0, linestyle="--", color="black")
plt.title("Question 6: Marginal Cost Changes")
plt.xlabel("Marginal Cost Change")
plt.ylabel("Count")
plt.show()
```

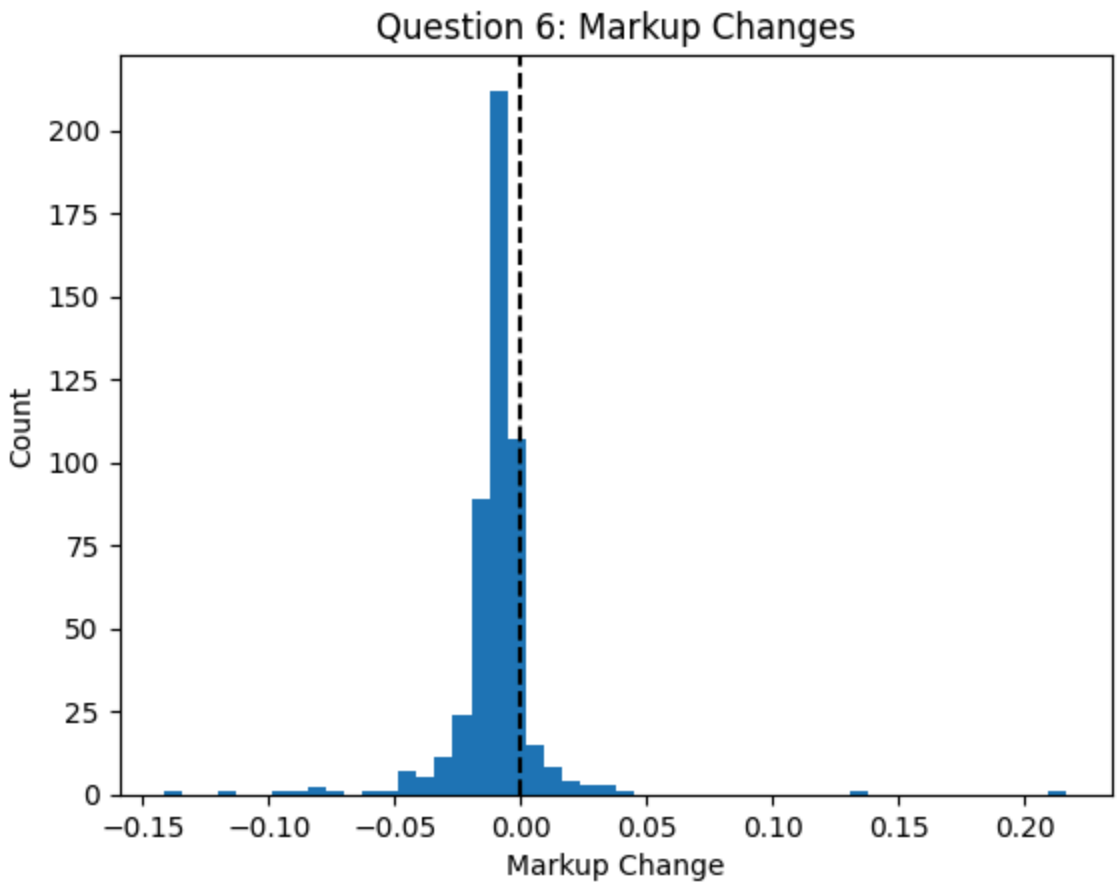
	delta_mc
count	5.000000e+02
mean	6.917854e-02
std	9.278372e-02
min	-5.305054e-10
25%	-3.882950e-12
50%	3.311424e-10
75%	1.308034e-01
max	2.992113e-01



6.3. New markups

In [30]:
```
markups_q6 = q6_simulation_results.compute_markups(costs=mc_q6).flatten()
markups_orig = results.compute_markups(costs=mc_orig).flatten()
markup_change = markups_q6 - markups_orig

display(pd.Series(markup_change).describe().to_frame("delta_markup"))
plt.hist(markup_change, bins=50)
plt.axvline(0, linestyle="--", color="black")
plt.title("Question 6: Markup Changes")
plt.xlabel("Markup Change")
plt.ylabel("Count")
plt.show()
```

	delta_markup
count	500.000000
mean	-0.009958
std	0.019265
min	-0.141346
25%	-0.013546
50%	-0.009433
75%	-0.004538
max	0.216549

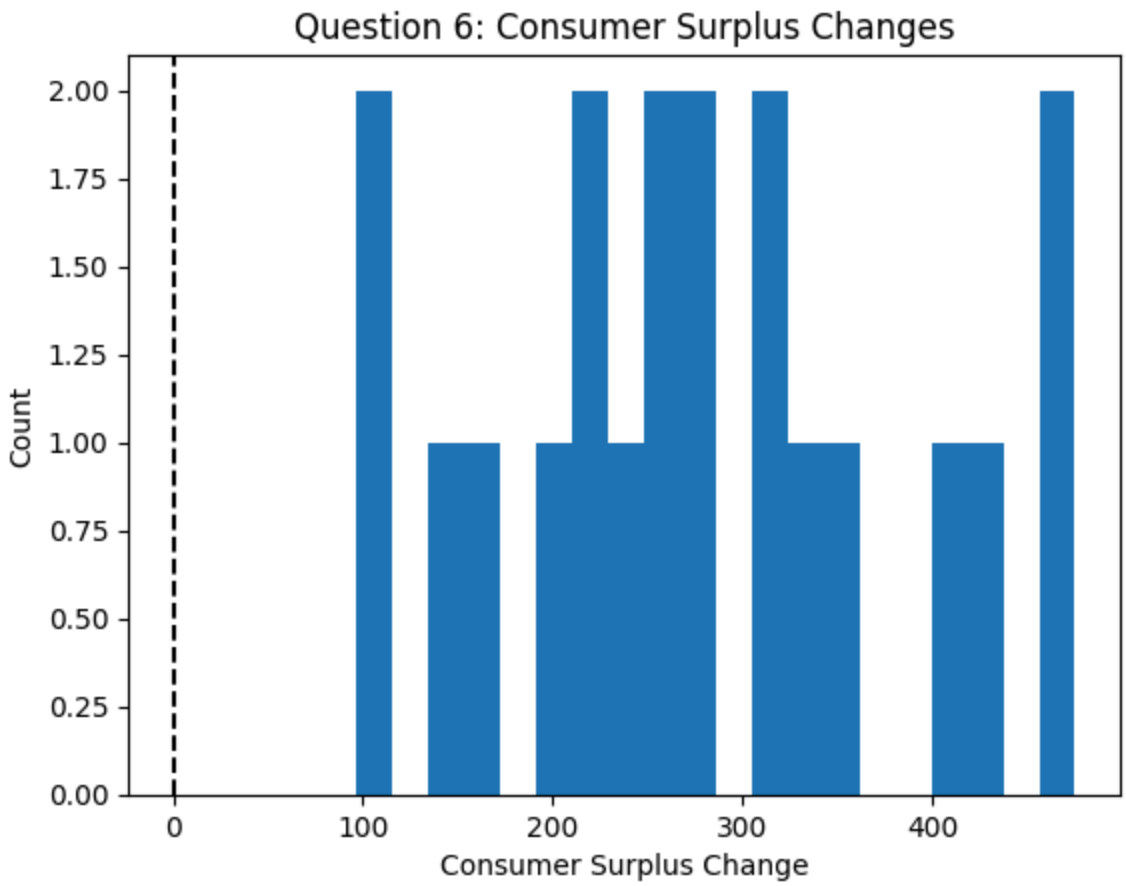


6.4. Consumer surplus

In [31]:
```
market_population = 100000 / product_data['market_ids'].nunique()
cs_q6 = q6_simulation_results.compute_consumer_surpluses().flatten() * marke
cs_orig = results.compute_consumer_surpluses().flatten() * market_populatior
cs_change = cs_q6 - cs_orig

q6_cs_table = pd.DataFrame({
    'market_ids': np.sort(product_data['market_ids'].unique()),
    'baseline_consumer_surplus': cs_orig,
    'new_consumer_surplus': cs_q6,
    'delta_consumer_surplus': cs_change,
})

display(q6_cs_table)
display(pd.Series(cs_change).describe().to_frame("delta_consumer_surplus"))
plt.hist(cs_change, bins=20)
plt.axvline(0, linestyle="--", color="black")
plt.title("Question 6: Consumer Surplus Changes")
plt.xlabel("Consumer Surplus Change")
plt.ylabel("Count")
plt.show()
```

	market_ids	baseline_consumer_surplus	new_consumer_surplus	delta_consumer_su
0	0	16080.389550	16300.644242	220.25
1	1	22013.496896	22122.108142	108.61
2	2	16301.487999	16453.975729	152.48
3	3	19814.244550	20028.080203	213.83
4	4	21668.800799	22023.849396	355.04
5	5	25058.239120	25370.223151	311.98
6	6	17304.326631	17552.933179	248.60
7	7	21138.102855	21545.446653	407.34
8	8	18188.094517	18425.445052	237.35
9	9	16944.826311	17208.696799	263.87
10	10	18760.393458	19179.293229	418.89
11	11	26202.900979	26533.266814	330.36
12	12	21262.578949	21737.649232	475.07
13	13	23381.521953	23664.898508	283.37
14	14	16232.831923	16329.088165	96.25
15	15	21066.658154	21372.596108	305.93
16	16	18797.798392	18960.042030	162.24
17	17	21610.866471	21880.883343	270.01
18	18	17758.509771	17963.788377	205.27
19	19	19436.140941	19904.628557	468.48

	delta_consumer_surplus
count	20.000000
mean	276.766334
std	109.715731
min	96.256242
25%	211.696391
50%	266.943680
75%	336.536525
max	475.070283



Interpretation

The simulation converged with 0 fixed-point failures, so we were able to solve for the new equilibrium after imposing the 33-minute delivery ceiling and holding ξ and ω fixed.

Prices move in both directions across products, but the average price change is positive. The mean change in price is about 0.08 dollars per slice, while the median is slightly negative at about -0.01 dollars. That suggests some products raise prices quite a bit after the cap, while many others change very little or even get slightly cheaper. A simple way to think about it is that the cap makes some pizzas more attractive to consumers, which pushes prices up, but it also raises marginal cost for the products whose delivery times are forced down, so the total price effect differs across products. Market-share changes are small on average, with a mean change of about -0.0004, but there is still variation across products.

Marginal costs increase on average. The mean change in marginal cost is about 0.07 dollars per slice, while the median is essentially 0. This fits the idea that the ceiling mainly matters for products that were originally above 33 minutes, since those are the products whose costs need to adjust.

Markups fall slightly on average. The mean change in markup is about -0.01 dollars per slice, and the median is about -0.009 dollars. So even though some firms benefit on the

demand side from faster delivery, the cost increase is a bit stronger on average, which leads markups to fall slightly overall.

Consumer surplus increases in every market. Since the assignment asks us to assume there are 100,000 consumers in total, consumer surplus is scaled by 5,000 consumers per market across the 20 markets. Under that scaling, the mean increase in consumer surplus is about 277 dollars per market, with the smallest increase around 96 dollars and the largest around 475 dollars. This suggests that consumers value the shorter delivery times enough that overall welfare rises in every market under the counterfactual.

Question 7 - Gentrification counterfactual

First we simulate the change in the agent distribution by changing the weights. For all rows where `invincome < 0.2`, we multiply the weights by 5 and do not redraw the population. Then we renormalize the weights within each market so that they sum to 1.

Holding the estimates from the original model fixed, we then simulate the new equilibrium and calculate the new prices, market shares, marginal costs, and markups.

In [32]:
```
agent_data.head()
```

Out[32]:

	v1	market_ids	nodes0	nodes1	weights	invincome
0	0	0	0.052184	-0.127630	0.001	0.693219
1	1	0	-2.037323	-1.195798	0.001	0.177587
2	2	0	0.741513	0.443902	0.001	0.200069
3	3	0	-0.610369	-0.417627	0.001	0.664243
4	4	0	0.374030	-2.591600	0.001	0.307349

In [33]:
```
# Count how many rows have invincome < 0.2
count = (agent_data['invincome'] < 0.2).sum()
print(f"Number of agents with invincome < 0.2: {count}")

# number of rows in the agent data frame
len(agent_data)
```

```
Number of agents with invincome < 0.2: 3932
```

Out[33]: 20000

In [34]:
```
# Impose change in agent data
# copy data set
agent_gen = agent_data.copy()

# Increase weights on high income individuals
high_income = agent_gen['invincome'] < 0.2
agent_gen.loc[high_income, 'weights'] *= 5
```

```
# Renormalize weights
agent_gen['weights'] = (
    agent_gen['weights'] /
    agent_gen.groupby('market_ids')['weights'].transform('sum')
)
```

In [35]:
```
# Keep residuals from earlier model
xi = results.xi
omega = results.omega
```

In [36]:
```
simulation = pyblp.Simulation(
    product_formulations = (
   pyblp.Formulation('0 + prices + deliverytime + rating'),
   pyblp.Formulation('1 + prices'),
   pyblp.Formulation('1 + rating + hourlywages + deliverytime')
),
        beta=results.beta,
        sigma=results.sigma,
        gamma=results.gamma,
        product_data=product_data,
        agent_data=agent_gen,
        agent_formulation=pyblp.Formulation('0 + invincome'),
        pi=results.pi,
        xi=xi,
        omega=omega,
        seed=1,
    )

simulation
```

Out[36]:
```
Dimensions:
============================================
 T    N    F     I     K1    K2    K3    D
---  ---  ---  -----  ----  ----  ----  ---
20   500   6   20000   3     2     4     1
============================================

Formulations:
==============================================================================
========
        Column Indices:              0           1             2
3
-------------------------------  ---------  ------------  -----------  ----
--------
  X1: Linear Characteristics      prices    deliverytime    rating
 X2: Nonlinear Characteristics       1         prices
X3: Linear Cost Characteristics      1         rating     hourlywages  deli
verytime
        d: Demographics          invincome
==============================================================================
========

Nonlinear Coefficient True Values:
==================================================
Sigma:     1       prices    |   Pi:    invincome
------  --------  --------   |  ------  ---------
  1     +1.3E-01             |    1     +0.0E+00
prices  +0.0E+00  +0.0E+00   |  prices  -4.1E+00
==================================================

Beta True Values:
================================
 prices   deliverytime   rating
--------  ------------  --------
-1.1E-02    -8.1E-01    +9.0E-01
================================

Gamma True Values:
=============================================
   1       rating   hourlywages  deliverytime
--------  --------  -----------  ------------
+6.9E-01  +4.7E-01   +1.4E+00      -6.7E-01
=============================================
```

In [37]:
```
# Replace simulated prices and market shares with equilibrium values that ar
simulation_results = simulation.replace_endogenous()
simulation_results
agent_data=simulation.agent_data

df= pd.DataFrame(pyblp.data_to_dict(agent_data))
```

In [38]:
```
# Convert solved simulation into a problem
problem = simulation_results.to_problem()

product_data_new=problem.products
```

```
df = pd.DataFrame(pyblp.data_to_dict(product_data_new))
```

7.1. Prices and Market Shares

In [39]:
```
# Build a product-level table comparing baseline and gentrification outcomes
q7_equilibrium = product_data[['v1', 'market_ids', 'firm_ids', 'prices', 'sh
q7_equilibrium = q7_equilibrium.rename(columns={'prices': 'baseline_price',
q7_equilibrium['new_price'] = df['prices'].to_numpy()
q7_equilibrium['new_share'] = df['shares'].to_numpy()
q7_equilibrium['delta_price'] = q7_equilibrium['new_price'] - q7_equilibrium
q7_equilibrium['delta_share'] = q7_equilibrium['new_share'] - q7_equilibrium

display(q7_equilibrium.head(10))
```

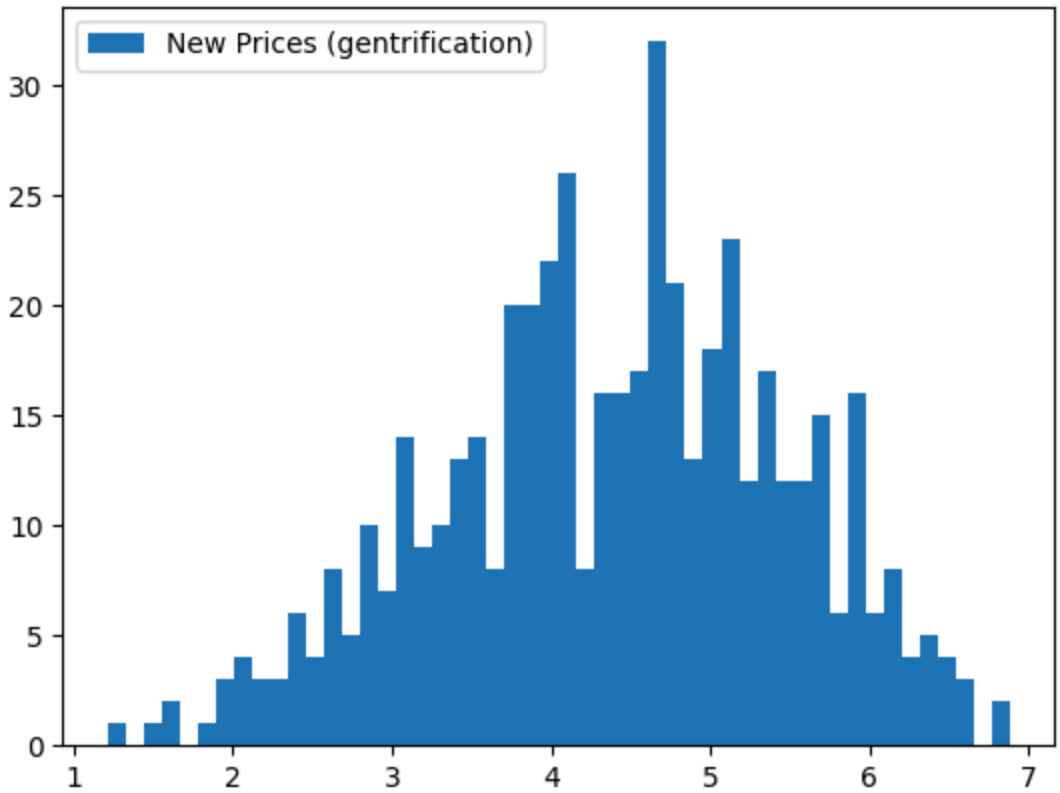
	v1	market_ids	firm_ids	baseline_price	baseline_share	new_price	new_share	del
0	0	0	0	4.586001	0.006038	5.528840	0.015255	0
1	1	0	0	4.171702	0.008047	5.295437	0.018054	
2	2	0	0	4.966030	0.004980	5.718871	0.013875	
3	3	0	0	4.120471	0.006846	5.264095	0.015120	
4	4	0	0	3.267534	0.011540	4.638194	0.019056	
5	5	0	1	0.968238	0.060462	1.592461	0.037837	0
6	6	0	1	4.306469	0.007761	5.297997	0.018618	
7	7	0	1	1.987181	0.050265	3.172252	0.048800	
8	8	0	1	4.546549	0.003045	5.443337	0.007780	0
9	9	0	2	4.438455	0.007630	5.388657	0.018897	0

Prices

In [40]:
```
# Summarize new prices and price changes across products.
display(pd.Series(df['prices']).describe().to_frame('new_price'))
price_change = df['prices'] - product_data['prices']
display(pd.Series(price_change).describe().to_frame('delta_price'))

plt.hist(df['prices'], bins=50)
plt.legend(["New Prices (gentrification)"])
plt.show()
```

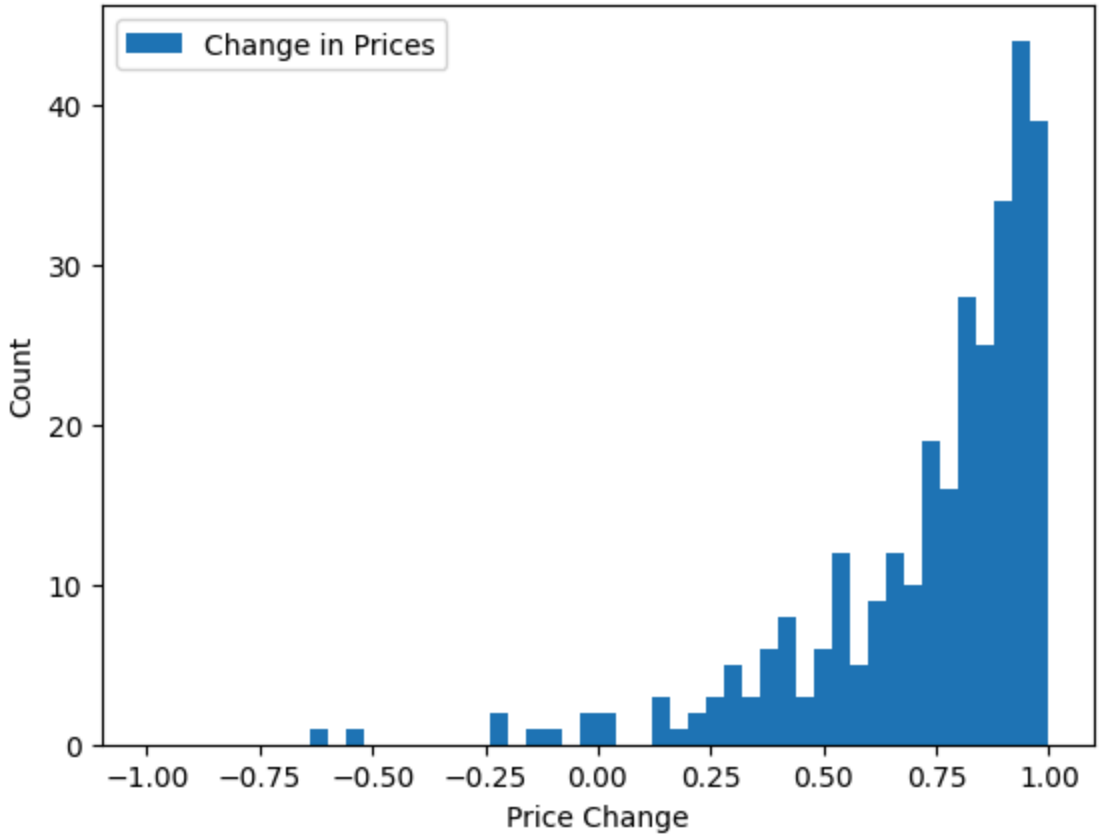
	new_price
count	500.000000
mean	4.391889
std	1.089332
min	1.213414
25%	3.695022
50%	4.441666
75%	5.160518
max	6.880629

	delta_price
count	500.000000
mean	0.892552
std	0.293310
min	-0.634514
25%	0.786577
50%	0.949524
75%	1.062549
max	1.518653



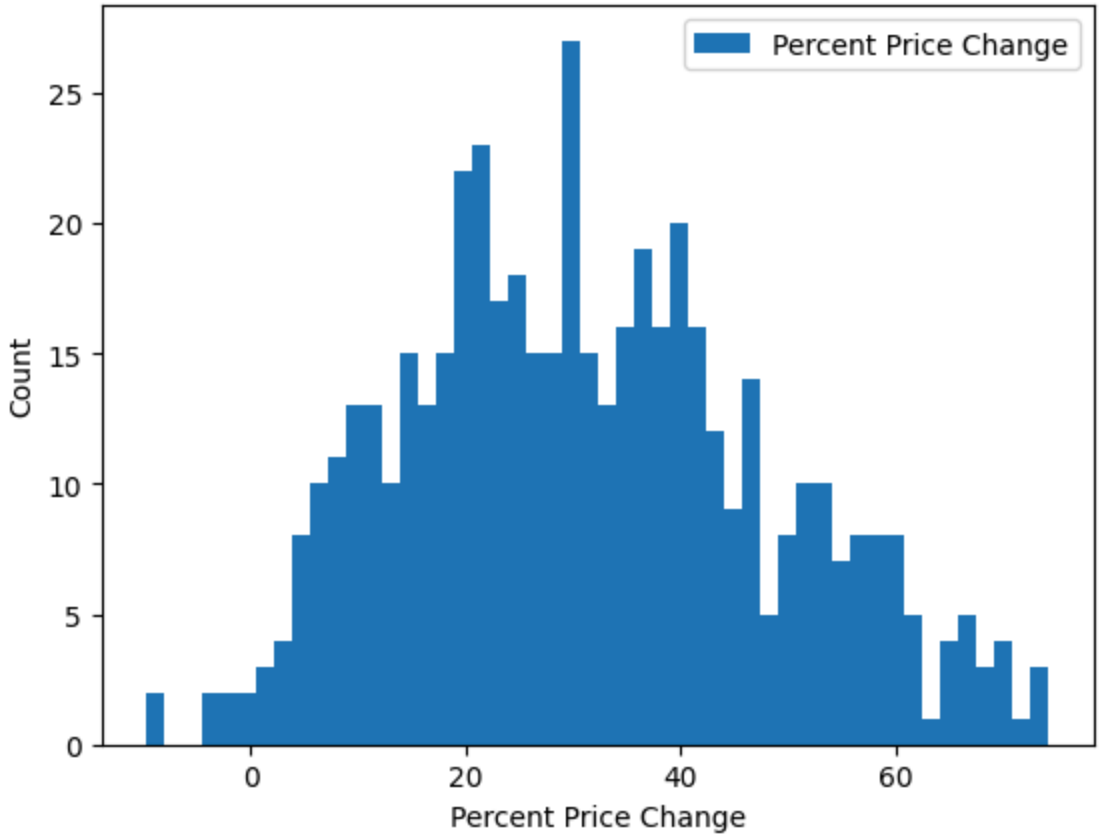
In [41]:
```
# Plot the distribution of price changes.
price_change = df['prices'] - product_data['prices']

plt.hist(price_change, bins=50, range=[-1,1])
plt.legend(["Change in Prices"])
plt.xlabel("Price Change")
plt.ylabel("Count")
```

Out[41]: Text(0, 0.5, 'Count')



In [42]:
```
# Percent changes in prices
price_pct_change = 100 * (df['prices'] / product_data['prices'] - 1)

plt.hist(price_pct_change, bins=50)
plt.xlabel("Percent Price Change")
plt.ylabel("Count")
plt.legend(["Percent Price Change"])
```

Out[42]:
```
<matplotlib.legend.Legend at 0x115ef6ad0>
```

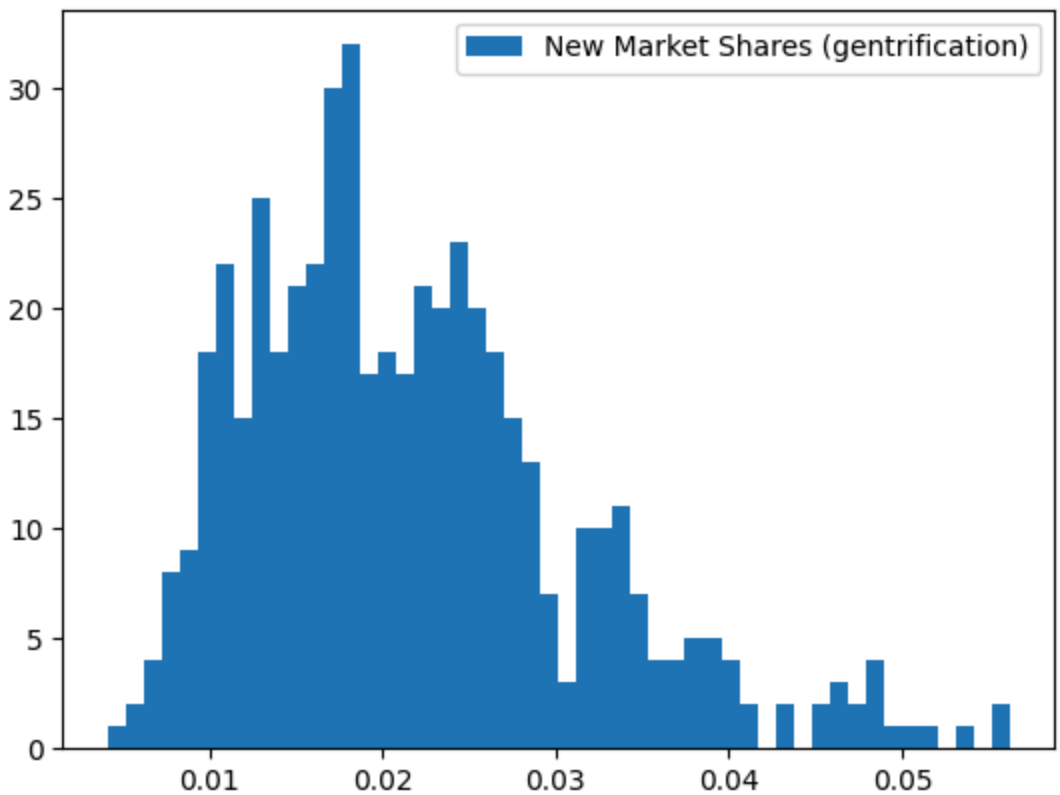


Market Shares

In [43]:
```
# Summarize new market shares and share changes across products.
display(pd.Series(df['shares']).describe().to_frame('new_share'))
share_change = df['shares'] - product_data['shares']
display(pd.Series(share_change).describe().to_frame('delta_share'))

plt.hist(df['shares'], bins=50)
plt.legend(["New Market Shares (gentrification)"])
plt.show()
```

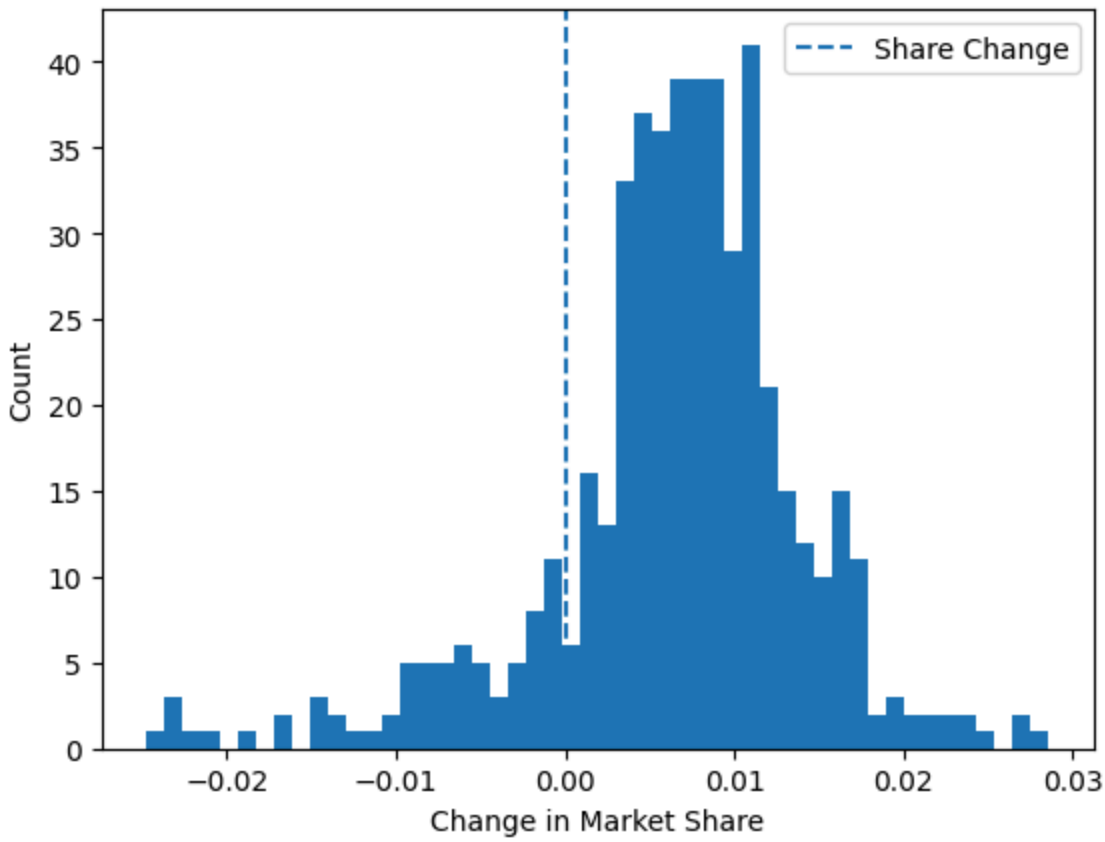
	new_share
count	500.000000
mean	0.021710
std	0.009545
min	0.004154
25%	0.014674
50%	0.020090
75%	0.026745
max	0.056210

	delta_share
count	500.000000
mean	0.006519
std	0.007701
min	-0.024705
25%	0.003590
50%	0.007243
75%	0.010856
max	0.028582



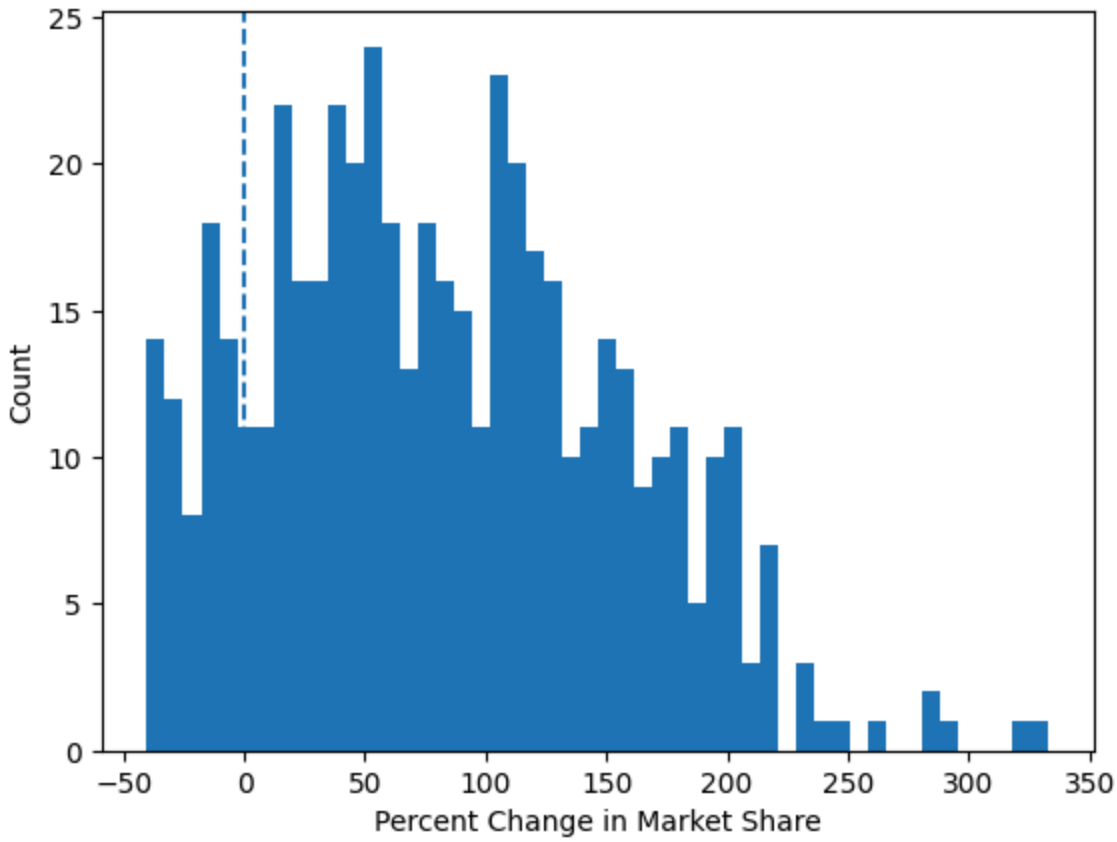
In [44]:
```
# Plot the distribution of share changes.
share_change = df['shares'] - product_data['shares']

plt.hist(share_change, bins=50)
plt.axvline(0, linestyle='--')
plt.xlabel("Change in Market Share")
plt.ylabel("Count")
plt.legend(["Share Change"])
```

Out[44]:
```
<matplotlib.legend.Legend at 0x1162f7610>
```



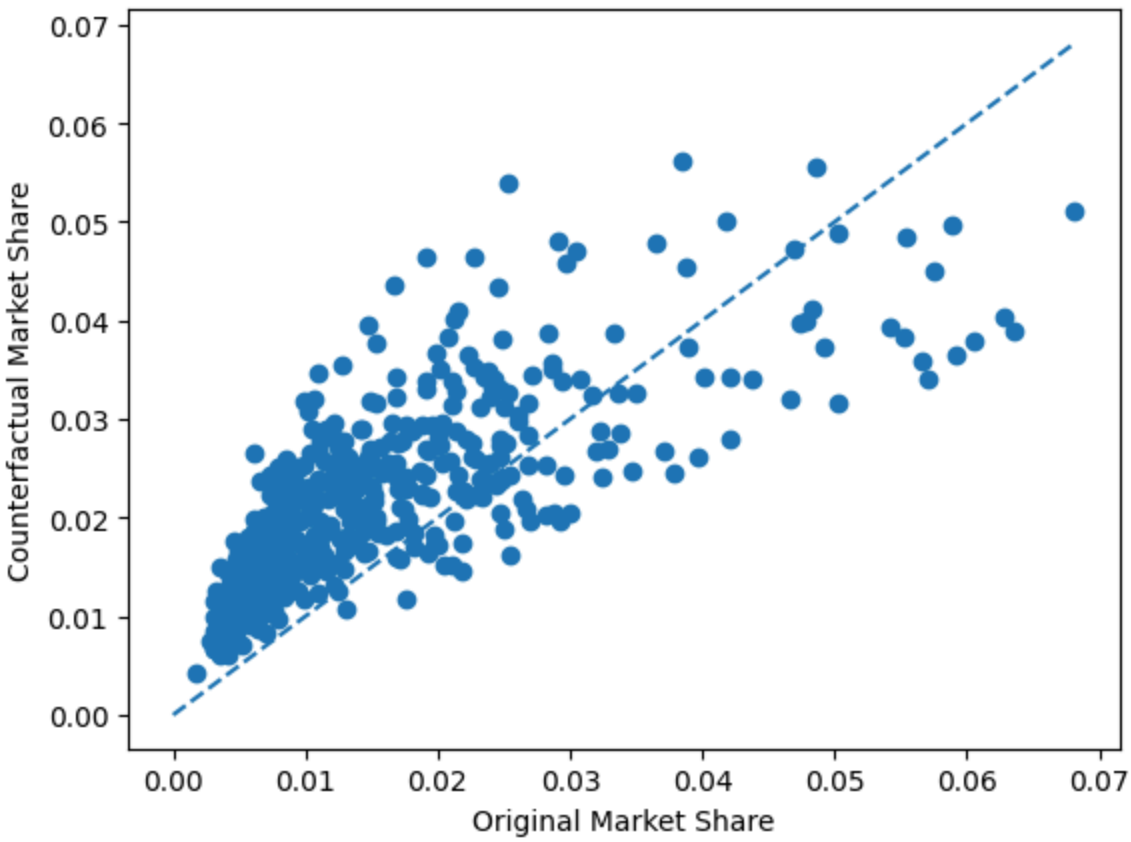
In [45]:
```
# Percent changes in market share
share_pct_change = 100 * (df['shares'] / product_data['shares'] - 1)

plt.hist(share_pct_change, bins=50)
plt.axvline(0, linestyle='--')
plt.xlabel("Percent Change in Market Share")
plt.ylabel("Count")
```

Out[45]:
```
Text(0, 0.5, 'Count')
```



In [46]:
```
# As a scatter plot
plt.scatter(product_data['shares'], df['shares'])
plt.plot([0, max(product_data['shares'])],
         [0, max(product_data['shares'])],
         linestyle='--')

plt.xlabel("Original Market Share")
plt.ylabel("Counterfactual Market Share")
```

Out[46]:
```
Text(0, 0.5, 'Counterfactual Market Share')
```



Interpretation

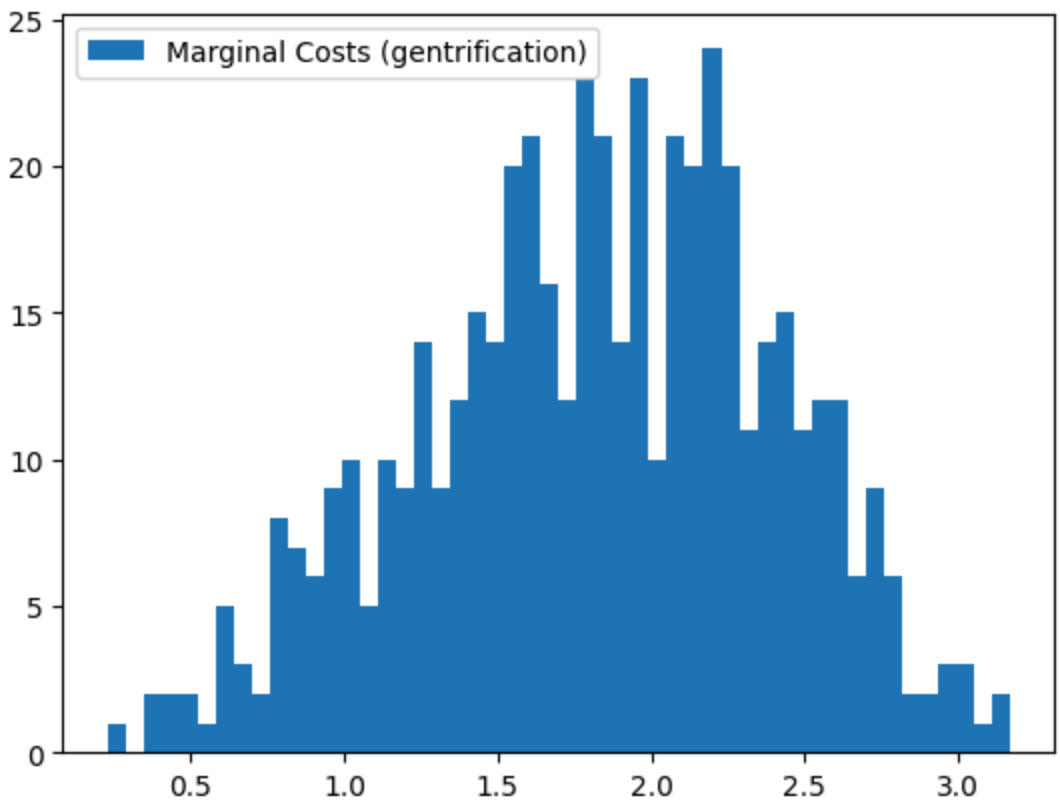
The gentrification counterfactual raises prices on average. The average price increase is about 0.893 dollars per slice, with most products showing positive price changes. This fits the idea that a larger share of higher-income consumers makes demand less price sensitive on average, so firms can profitably charge higher prices.

Market shares also increase on average, by about 0.0065, but the distribution is mixed across products. Some pizzas gain a lot of share while others lose share, which makes sense because the change in the income distribution affects substitution patterns across products rather than shifting every product in exactly the same way.

Marginal costs

In [47]:
```
# Summarize new marginal costs and their changes.
mc_gen = simulation_results.compute_costs().flatten()
display(pd.Series(mc_gen).describe().to_frame('new_marginal_cost'))

plt.hist(mc_gen, bins=50)
plt.legend(["Marginal Costs (gentrification)"])
plt.show()
```

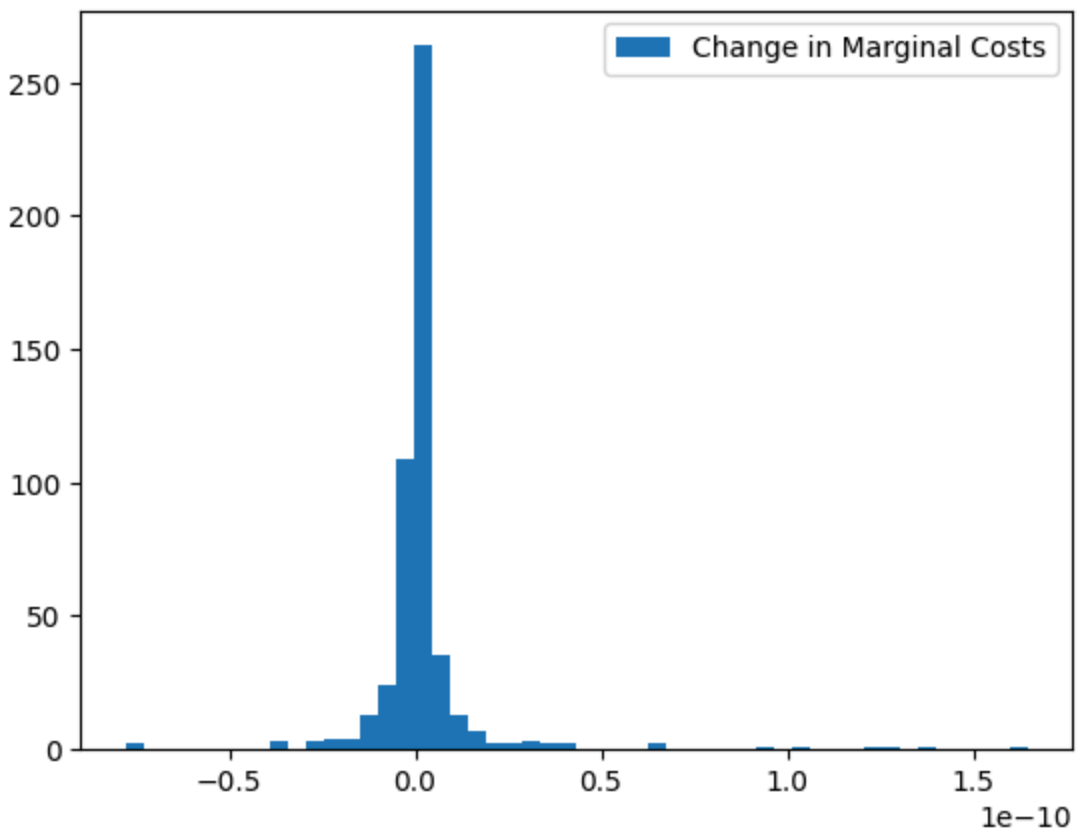
	new_marginal_cost
count	500.000000
mean	1.823218
std	0.578773
min	0.230516
25%	1.429346
50%	1.845768
75%	2.246162
max	3.166607



In [48]:
```
# Summarize and plot marginal-cost changes.
mc_orig = results.compute_costs().flatten()
mc_change = mc_gen - mc_orig
display(pd.Series(mc_change).describe().to_frame('delta_marginal_cost'))

plt.hist(mc_change, bins=50)
plt.legend(["Change in Marginal Costs"])
```

	delta_marginal_cost
count	5.000000e+02
mean	2.140064e-12
std	1.706366e-11
min	-7.787682e-11
25%	-1.318279e-12
50%	8.223422e-13
75%	2.547740e-12
max	1.645115e-10

Out[48]: <matplotlib.legend.Legend at 0x1164dd810>

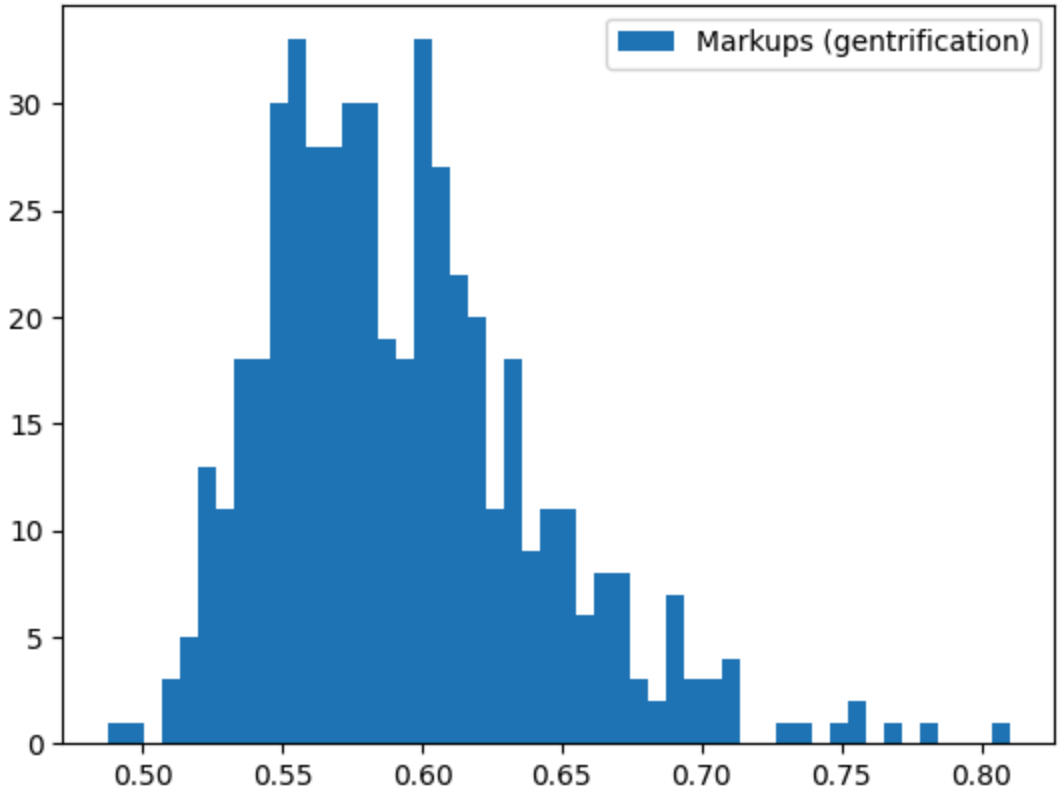


Interpretation

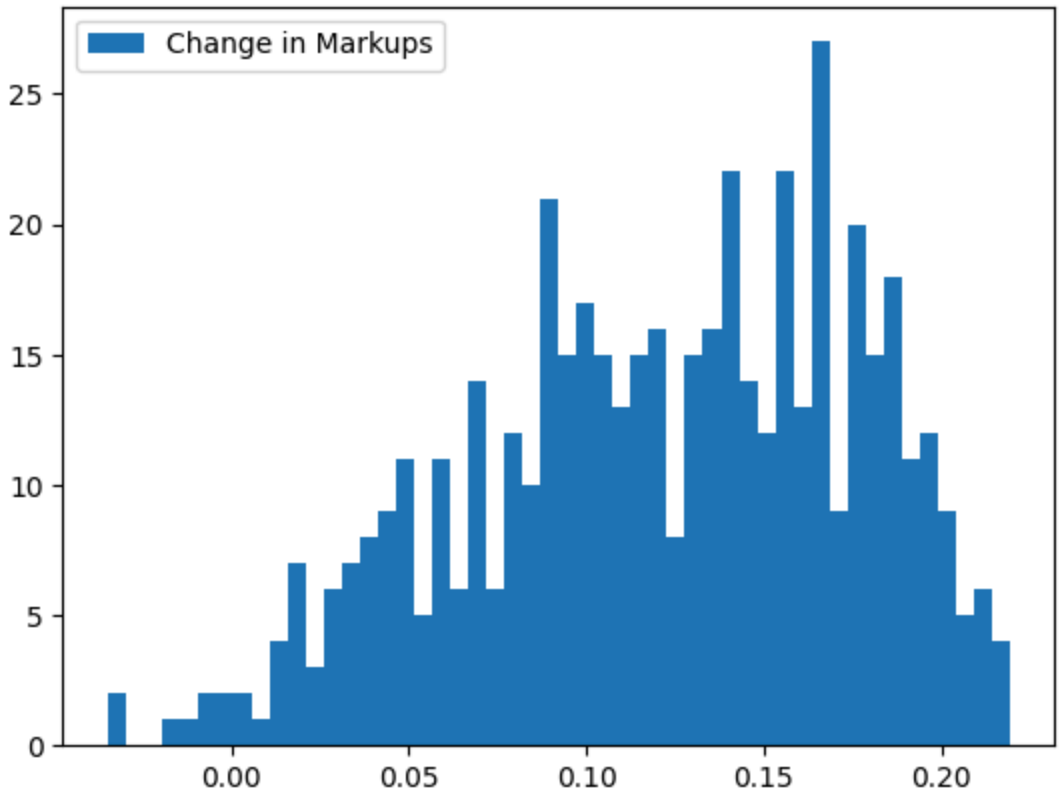
Marginal costs are essentially unchanged in this counterfactual. The average change in marginal cost is effectively 0, which makes sense because the exercise changes the composition of consumers on the demand side, not the underlying cost shifters on the supply side.

Markups

In [49]:
```
# Construct new markups
markups_gen = simulation_results.compute_markups(costs=mc_gen).flatten()
plt.hist(markups_gen, bins=50);
plt.legend(["Markups (gentrification)"]);
```



In [50]:
```
# Changes in markups
# Construct new markups
markups_gen = simulation_results.compute_markups(costs=mc_gen).flatten()

# Original markups
markups_orig = results.compute_markups(costs=mc_orig).flatten()

# Change in markups
mu_change = markups_gen - markups_orig

plt.hist(mu_change, bins=50);
plt.legend(["Change in Markups"]);
```



In [51]:
```
pd.Series(mu_change).describe()
```

Out[51]:
```
count    500.000000
mean       0.121163
std        0.054123
min       -0.034724
25%        0.084702
50%        0.127515
75%        0.165031
max        0.219198
dtype: float64
```

Interpretation

Markups increase in this counterfactual. The average markup increase is about 0.121 across products. Since prices rise while marginal costs stay close to their original levels, most of the increase in price shows up as higher markups rather than higher production costs.